Cross-Dataset Reliability and Computational Trade-offs in Cascaded URL Phishing Detection

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Dedication

I want to express my appreciation to my family and everyone who has supported me along the way. I would like to thank the countless mentors, teachers and managers throughout my career that supported me – even though they didn't see everything I had accomplished at that time. They saw the potential I had, and they believed I would succeed. I value all those experiences to this day. I also wish to recognize those people and organizations that have worked behind the scenes to make sure our critical systems continue to operate smoothly and safely.

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A number of security practitioners and colleagues from within research and from within organizations have discussed their experiences with detection in production including deployment of detection systems, their experiences of incident response and the limitations of detection in production that I have attempted to describe in this work – including colleagues in the enterprise. Additionally, I would like to thank all publicly available sources of data and/or benchmarks. Last but not least, I want to thank my family and friends, as well as online communities where I wrote, re-ran experiments, and discussed various aspects of this work in considerable detail while the rest of my life went on elsewhere.

Abstract of Praxis

Cross-Dataset Reliability and Computational Trade-offs in Cascaded URL Phishing Detection

This dissertation evaluates how URL representation, selective model execution and HTTP service design affect phishing detection under a low false-positive constraint. The CyberSentinel gateway was evaluated on 34,593 internal records, 8,701 external records and 125 operational cells. Structural features increased internal recall by 64.07 percentage points over length alone. Eight of 22 original component checks pass after correction of paired uncertainty; H1–H3 remain unsupported because their joint detection, monitoring and service requirements are not met. Mechanism-directed comparisons explain these outcomes. A frozen-model scheme intervention reduced certified false positives by 2,211 but lost 47 of 69 gold detections, showing why formatting cannot be repaired by optimizing false positives alone. A separately fitted scheme-neutral representation achieved exact feature, score and decision invariance across 8,622 companion pairs, with a measured recall–specificity tradeoff. A post-review, domain-balanced retrospective cohort contained 1,392 positive and 1,262 negative domains. Both the original and revised cascades detected all positives; their corrected 95% recall-difference bounds establish recall-only noninferiority to the transformer under independent labeled-pair assumptions. The revised symmetric gate nevertheless added 18 false positives without recovering a missed positive. On a separate fixed workload of real URL strings, it required 1.646 times the structural scorer's summed processing time. Five new shuffled-label fits retain direct evidence of the labels consumed by each estimator. Separately, worker-owned HTTP clients reduced median paired structural-workload p95

by 79.68%, with zero errors in 100,000 primary worker requests; strict response equivalence was not achieved. The contribution is a measured account of conditional expert repair, representation stability and service cost that identifies which design changes improve a property and which fail the joint operating requirements.

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List of Symbols

TP, FP, TN, FN — True-positive, false-positive, true-negative and false-negative counts against source reference labels.

P, N — Positive and negative stratum sizes in confusion-count tables; a total stream size is identified explicitly where used.

X; P(X) — Observed feature vector; its marginal distribution.

τ — Validation-selected decision threshold, unchanged during evaluation.

p50, p95, p99 — 50th, 95th and 99th percentiles of individual measured terminal request latencies.

pp — Percentage points; the unit of differences between proportions expressed as percentages.

c — HTTP client concurrency, such as c64 for 64 concurrent clients.

List of Acronyms

AC — Alternating-current power; the monitored external-power condition.

AI — Artificial intelligence

AP — Average precision

API — Application programming interface

ASCII — American Standard Code for Information Interchange.

ASGI — Asynchronous Server Gateway Interface

AUC — Area under the curve

BIC — Bayesian information criterion

CI — Continuous integration in the software-verification record; uncertainty intervals are written out in the manuscript.

CLI — Command-line interface.

CP — Clopper–Pearson

CPU — Central processing unit

CSV — Comma-separated values

DNS — Domain Name System

DOI — Digital object identifier.

ECE — Expected calibration error

FIFO — First-in, first-out queue discipline.

FNR — False-negative rate

FPR — False-positive rate

GB — Gigabytes.

GELU — Gaussian error linear unit.

GMM — Gaussian mixture model

HTTP — Hypertext Transfer Protocol

HTTPS — Hypertext Transfer Protocol Secure

IDNA — Internationalizing Domain Names in Applications.

JSON — JavaScript Object Notation

MCC — Matthews correlation coefficient

MMD — Maximum mean discrepancy

MPS — Metal Performance Shaders, the Apple transformer-execution backend.

NCSC — National Cyber Security Center (source designation in PhishVN)

OS — Operating system.

PCG64 — NumPy's 64-bit-output permuted congruential generator.

PSI — Population stability index

RF — Random Forest

RFC — Request for Comments; RFC 3986 specifies URI syntax.

ROC — Receiver operating characteristic

SHA-256 — Secure Hash Algorithm with a 256-bit digest.

TPR — True-positive rate

URL — Uniform resource locator

UTF-8 — Unicode Transformation Format, 8-bit encoding.

Glossary of Terms

Admission sequence — Service-assigned order of admission. It is part of exact response agreement, not a predictive field.

Calibration — Agreement between stated predictive probabilities and observed reference-label frequencies; distinct from ranking and threshold performance.

Closed-loop timing — A workload in which each client starts another request after its previous request completes. It does not represent an unrestricted arrival process.

Clustered bootstrap — Resampling whole groups, such as registrable domains, rather than treating all records as independent.

Discrimination — Ability to rank positive reference-label cases above negative cases; distinct from calibration.

Frozen threshold — A decision boundary selected using development evidence and kept unchanged for the specified evaluation.

Invocation — An actual execution of a model, rather than a routing intention or an accepted HTTP request.

Metamorphic relation — A declared relation between transformed inputs and expected outputs, used to test a specific behavioral property.

Noninferiority margin — The prespecified maximum acceptable loss relative to a comparator for a defined outcome and uncertainty rule.

Paired run — Two comparator executions matched by experimental condition and pair identifier. Run-pair summaries, not individual requests, are resampled for the service-ratio interval.

Registrable domain — The public suffix plus the immediately preceding domain label, determined under the retained suffix rules and used as the grouping key.

Source shift — A change in collection source or its resulting population. It does not by itself establish chronological drift or changed prediction errors.

Tail latency — A high percentile of the measured request-latency distribution, such as p95 or p99; its success/error denominator must be specified.

Transport-neutral representation — The declared feature representation that removes sensitivity to the prescribed HTTP/HTTPS scheme-only transformation; it does not establish equivalence of live pages.

Chapter 1—Introduction

1.1 Background

Applications receive and produce URLs through APIs, retrieval pipelines and generated responses. A pre-fetch gateway can inspect those strings before they reach a user or another system. CyberSentinel evaluates this narrow control without fetching destinations, rendering pages or querying DNS, WHOIS or reputation services. Its information is restricted to URL syntax, structure and character sequence.

Two tensions motivate the investigation. Related domains can inflate apparent generalization when their URLs cross training and test partitions. Richer character representations may recover patterns missed by a structural model, but their computation is useful only if it improves decisions at an acceptable false-positive rate. Domain-separated evaluation, selective execution and distribution monitoring address these concerns; the study tests whether their benefits survive common detection and service constraints.

1.2 Problem Statement

An inline detector needs more than aggregate accuracy. It must preserve specificity on external sources, identify where an expensive expert improves a cheaper model's decisions, and meet latency and error budgets through the complete HTTP path. A favorable result on one dimension cannot establish the others.

Public labels reflect corpus collection and reference classification, not necessarily current forensic truth. Mechanical mapping, quarantine and domain allocation therefore precede fitting and evaluation. Staged freezes separate those

decisions from model, threshold and routing selection. Later changes to publisher-normalized inputs and two-session execution have distinct information boundaries, identified in Sections 3.4, 3.10 and 4.6. The analysis preserves those boundaries rather than presenting every decision as prediction-blind.

1.3 Purpose of the Study

CyberSentinel combines L1-regularized structural logistic regression with a compact character transformer. A validation-selected uncertainty band determines selective execution. A GMM monitors structural features and the first-stage score; in ordered offline replay, its alerts change routing for the next 256 requests. This future-only policy is not prospective field collection.

PhiUSIIL supplies development and domain-disjoint internal evaluation; PhishVN v3.1.0, Mendeley repository Version 4, supplies external evaluation. The completed original study covers 22 checks and 125 operational cells in 25 groups. Eight checks pass after paired-inference correction, without changing any numerical requirement. Chapters 4–5 explain the resulting H1–H3 decisions and examine ranking, calibration, comparators and robustness.

Mechanism-directed comparisons extend that evaluation. D tests a refitted scheme-neutral representation; S changes client connection ownership. Post-review F isolates scheme spelling under fixed weights, R tests selected routing rules on previously unscored domains, C measures actual detector computation, and P audits five shuffled-label fits. Their separately specified outcomes explain the original results rather than replace them.

1.3.1 Thesis Statement

CyberSentinel will test low-FPR structural and character-level detection, GMM-guided escalation, and whether selective execution stays within two percentage points of transformer-only recall while invoking the transformer no more than 30%, keeping p95 at or below 200 ms, and holding errors below 0.1%.

The preceding statement preserves the working thesis on slide 5 of the August 20 advisor deck. Its future tense records the commitment at that stage of the project; it is not a statement of achieved performance. The three research questions and their conjunctive hypotheses operationalize that commitment. Chapter 4 reports each decision, and Chapter 5 explains the relationship between the demonstrated component benefits and the joint inline operating claim.

1.3.2 Research Objectives

The first objective is to quantify the incremental recall and false-positive behavior of structural and character representations on registrable-domain-disjoint internal data and external source/tier strata. The second is to measure distribution sensitivity, reference false alerts and the consequences of future-only escalation without treating an alert as proof of harmful drift. The third is to measure actual HTTP latency, throughput, transformer attempts and request errors against the frozen operational constraints. Together these objectives provide an end-to-end answer about the tested system, rather than separate favorable accuracy and speed claims.

1.4 Research Questions and Hypotheses

The National Cyber Security Center (NCSC) source designation below is the publisher's label for a PhishVN positive stratum, not an independent adjudication performed by this study.

1.4.1 RQ1 and H1

RQ1: What incremental value do structural URL features and character-level representations provide under registrable-domain-disjoint and external evaluation?

H1: At the validation-selected FPR ceiling, full structural features improve recall over a length-only baseline, and the character-transformer cascade improves recall over Logistic-L1.

H1 is supported only if each of length-only, Logistic-L1, and cascade has observed FPR <= 1% on both PhiUSIIL group-test negatives and certified trusted-registry negatives, and if the registrable-domain-clustered 95% bootstrap lower bounds exceed zero for all four prespecified recall differences. The differences are Logistic-L1 minus length-only and cascade minus Logistic-L1, each evaluated on PhiUSIIL group-test positives and NCSC gold positives. A positive difference in only one source, or an improvement accompanied by excess FPR, does not support H1.

These are the only two primary H1 contrasts. Transformer-only remains a comparator and operational reference, not a third primary H1 gate. The cascade-minus-Logistic-L1 contrast measures the selective system contribution, not a pure causal isolation of representation.

1.4.2 RQ2 and H2

RQ2: Can GMM-based monitoring detect an external source/domain shift and guide escalation without exceeding the low-FPR operating constraint?

H2: GMM detects at least 80% of prespecified external shift windows at no more than 5% false alerts, and prospective routing reduces the false-negative rate relative to the fixed cascade while retaining FPR <= 1%.

H2 is supported only if all four conditions hold: external-window detection is at least 80%; false alerts on the independent PhiUSIIL validation audit stream are no more than 5%; the alert policy's observed FPR on certified trusted-registry negatives is <= 1%; and the lower bound of the registrable-domain-clustered 95% bootstrap interval for false-negative-rate reduction on NCSC gold positives is greater than zero. The monitor measures departure in P(X). It does not by itself establish harmful drift, label shift, concept drift, or a causal change in model error.

1.4.3 RQ3 and H3

RQ3: What detection, escalation, throughput, and latency tradeoffs determine whether the fixed cascade is viable inline?

H3: For each system, observed FPR on certified trusted-registry negatives is <= 1%, and the Tranco reference-negative alert rate is <= 1% as a mandatory secondary safeguard. Cascade recall on NCSC gold positives is noninferior to transformer-only recall within 2 percentage points. On the 1%-prevalence benchmark reference, transformer invocation is at most 30%, real-HTTP p95 latency is at most 200 ms at concurrency 64, and the request-error rate is below 0.1%.

H3 requires both external-FPR safeguards for both the fixed cascade and transformer-only model. Exact one-sided 95% Clopper-Pearson upper confidence bounds accompany the two certified-registry FPR estimates and the two Tranco control alert-rate estimates. The domain-clustered 95% bootstrap lower bound for recall(cascade) minus recall(transformer-only) on NCSC gold positives must be at least -0.02. Stage 2 must be invoked for no more than 30% of the primary reference manifest. Pooled p95 latency across the measured concurrency-64 runs must be <= 200 ms, and the request-error numerator divided by all 50,000 measured concurrency-64 requests must be < 0.1%.

Sections 1.4.1–1.4.3 preserve the original research commitments, including their historical bootstrap specification. The post-review correction in Section 3.8 replaces inadequate paired uncertainty without relaxing any metric or numerical margin. Tables 4.3 and 4.14 and the current RQ/H answers use that correction; original estimates and decisions remain available for comparison in the retained record.

1.5 Contribution Boundary

CyberSentinel brings structural and character inference, future-only GMM-guided routing and real HTTP execution into one evaluated URL-only gateway. The study compares their behavior on domain-separated outcome strata under common operating requirements. It asks whether gains from an individual component remain useful when external false-positive risk, escalation cost, tail latency and reliability are considered together. The contribution is the implemented gateway and this connected evaluation of its components, rather than a new classifier architecture.

The subsequent comparisons add mechanism-level evidence to that systems account. Consistent scheme neutralization removes the prescribed HTTP/HTTPS

sensitivity from the structural representation, while an additional benchmark measures the detection tradeoff at frozen operating points. A paired client-topology experiment, including a no-model control, demonstrates a median paired p95 reduction of 79.68% for the structural workload at concurrency 64. Worker-owned clients meet that comparison's latency and error requirements, although exact response equivalence remains unmet because admission sequence is part of the specified contract. These results establish concrete improvements without treating representation conformance, external specificity and service correctness as interchangeable outcomes.

Prior work establishes structural and character URL models, domain-aware evaluation, selective cascades, source-shift analysis and security monitoring (Ahamed et al., 2026; Hussain et al., 2027; Li et al., 2021; Rashid et al., 2024; Tsai et al., 2024; Yang et al., 2021). This study does not claim invention of those components or of scheme normalization and connection reuse. Its technical value is the specific integrated artifact, controlled comparisons and traceable evidence that connect a diagnosed behavior to an implemented change and a measured engineering decision.

1.6 Scope, Delimitations, and Limitations

The detection unit of analysis is one retained URL record. The system produces allow or alert decisions; it does not automatically block requests, remove content or take down destinations. Only absolute HTTP and Hypertext Transfer Protocol Secure (HTTPS) URLs satisfying the declared parsing rules enter the retained development partitions. Here, raw-URL or URL-only detection identifies the information modality: the model receives a URL string without fetching its destination. It does not imply that every source supplies an untouched original spelling. The amended external evaluation uses the

publisher's exact url_norm, as specified in Section 3.4. Private request content, rendered pages, DNS, WHOIS, reputation services and network telemetry are outside the study. Frontier-AI inference motivates the gateway boundary; the experiments evaluate the URL service, not a frontier model's reasoning, browsing behavior or field deployment.

The primary label limitation is explicit. PhiUSIIL and the external source provide publisher-provided, source-derived reference classifications, not independently verified ground truth. PhiUSIIL does not expose public per-row source, timestamp, snapshot, or independent-adjudication fields. This limits record-level forensic verification and temporal analysis. The study addresses the limitation through exact source preservation, mechanical mapping, quarantine, source/domain-separated evaluation, and bounded claims. Those controls improve reproducibility; they do not transform reference labels into independently proven facts. External parsing and model inputs use the publisher's exact url_norm under the disclosed amendment, with original raw cells preserved. Publisher-added schemes and domain-only records cannot establish the transport or page behavior of a live destination.

All numerical targets are study-defined operating requirements, not thresholds prescribed by the literature. They include the <= 1% FPR ceiling, >= 80% shift-window detection, <= 5% false alerts, -0.02 recall noninferiority margin, <= 30% transformer invocation, <= 200 ms real-HTTP p95 latency, < 0.1% request errors and 2,000 ms client timeout. These requirements were fixed before confirmatory outcomes were known and evaluated without outcome-driven relaxation.

Chapter 2 relates the investigation to phishing detection, selective inference and distribution monitoring. Chapter 3 specifies the original research commitments, the

corrected inference, and the bounded follow-up comparisons. Chapter 4 presents the original results, representation and service interventions, and post-review diagnostic, routing and computation findings. Chapter 5 answers RQ1–RQ3, explains the engineering contribution and separates conclusions supported by the measurements from questions requiring another study. Appendices provide the reproduction map, complete service schedule and methodological chronology.

Chapter 2—Review of the Literature

This focused, problem-driven review examines six concerns that shape the inference-gateway design: raw uniform resource locator (URL) classification, structural and character representations, domain and source shift, low-false-positive-rate (FPR) operating points, selective execution and security drift monitoring. It compares the inputs, evaluation populations, decision rules and operational measurements used in prior work. These differences determine which findings can inform the present system. The review is not systematic.

This synthesis situates both the original investigation and its subsequent engineering comparisons. It does not imply that every cited publication informed the original freeze. Chapter 3 records when the development decisions and amendments were made; the literature supplies context and comparison, not retrospective prespecification.

2.1 Raw-URL Phishing Detection

Phishing detection can draw from page content, visual appearance, host infrastructure, reputation, email context, or the URL itself (Basit et al., 2021; Khonji et al., 2013). A raw-URL detector accepts less information than a full web classifier, but it can act before a page is fetched and can be applied consistently at an application programming interface (API) boundary. Lexical and structural indicators commonly include URL length, hostname and path lengths, punctuation counts, digit ratios, entropy, IP-literal use, port syntax, and suspicious authority forms. These variables are inexpensive and interpretable, but their usefulness depends on the source and period from which examples were collected.

PhiUSIIL was published as a large URL dataset assembled from phishing sources and an Open PageRank legitimate source (Prasad & Chandra, 2024a, 2024b). The associated work combined URL and webpage features. This study uses only the raw URL and features computed locally from it. That distinction matters because reported performance based on page-derived variables cannot be treated as an achieved raw-URL result. It also matters for labeling: the corpus's native class is preserved as a publisher reference rather than re-created from locally chosen heuristics.

The scope of the available signal is a central comparison dimension. Khonji et al. (2013) place detection within a wider set of phishing mitigations, including prevention and corrective action. Basit et al. (2021) review machine-learning, deep-learning, hybrid and scenario-based detection approaches. These surveys establish that a detector is one intervention in a broader security process. A URL decision alone cannot authenticate the operator of a page, establish the intention of a message sender or prove that a user will respond safely to a warning. The present gateway consequently evaluates URL classification and service behavior rather than claiming prevention of all phishing harm.

Earlier URL-learning work also differs in what it permits the model to observe. Ma et al. (2009) combine lexical and host-based properties to classify malicious web addresses; their task is broader than phishing alone. Sahingoz et al. (2019) compare classifiers using natural-language-processing-based URL features and emphasize independence from third-party services. Together, these studies establish learning from address structure as prior art, while distinguishing locally computed string features from infrastructure lookups. They support the feasibility of a lightweight first stage, not the assumption that its error rate will remain unchanged under another collection source.

Client-side execution is not synonymous with raw-URL-only detection. Jain and Gupta (2018) extract features from both the URL and page source without third-party dependencies. Rao and Pais (2019) combine URL, source-code and third-party features. Marchal et al. (2017) implement Off-the-Hook as a browser add-on and evaluate its warnings with users. These approaches can exploit evidence unavailable before page retrieval, or assess human outcomes absent from an inference endpoint. Comparing their headline accuracies directly with the present raw-string detector would confound feature access, population, labels and evaluation protocol. Their more useful contribution to this study is a set of design alternatives: richer evidence can justify additional fetching and interaction costs, whereas a pre-fetch gateway must demonstrate the value attainable from the restricted input it actually receives.

2.2 Structural and Character Representations

L1 regularization can suppress weak or redundant predictors, making a sparse structural model inspectable (Tibshirani, 1996). Random forests instead learn nonlinear combinations of supplied features through randomized trees (Breiman, 2001). Character encoders learn sequence information: Zhang et al. (2015) study character CNNs; URLNet combines character- and word-level URL representations (Le et al., 2018). Vaswani et al. (2017) introduce attention-based sequence modeling. These provide different capacities, not an assumed progression from simpler to better detectors.

Ahamed et al. (2026) evaluate structural and character models under adversarial, domain-disjoint and external conditions. Their PhishTank check includes held-out PhiUSIIL benign examples; a separate PhishTank/Majestic check uses an independent negative source. Their work establishes relevant combined-model and transfer evaluation,

rather than a gap that can be filled merely by adding a transformer. Here RQ1 tests incremental recall at a common low-FPR operating constraint across specified populations.

Scikit-learn provides consistent classical-model interfaces (Pedregosa et al., 2011). Implementation provenance still requires feature order, fitted state, versions and partitions, specified in Chapter 3. A richer model or a correct library call becomes useful only through measured out-of-sample decisions at the required operating point.

2.3 Domain Leakage and Source Shift

Related-domain URLs crossing training and test partitions can exaggerate generalization. Blocked-validation research explains both this dependence problem and the changed prediction task introduced by grouping (Roberts et al., 2017). Label-blind registrable-domain allocation controls that leakage route, not every source effect. Rashid et al. (2024) study cross-dataset phishing degradation and adaptation; Tsai et al. (2024) examine malicious-URL dataset bias and invariant representations.

PhishVN supplies source and confidence tiers with grouped splits (Vu, 2026a). Gold positives and certified trusted-registry negatives define primary external strata after label-blind routing. Tranco remains a popularity-based control: ranking composition and manipulation affect its use in security research (Le Pochat et al., 2019). Corpus identity and popularity cannot substitute for label authority.

Pan and Yang (2010) distinguish transfer settings. Ben-David et al. (2010) connect target error to source error and distribution divergence under shared-hypothesis assumptions. Sugiyama et al. (2007) study importance weighting under covariate shift, where $P(X)$ changes but $P(Y|X)$ does not. Changes in collection and labeling may violate

that assumption. The present frozen-model evaluation estimates neither their theoretical bound nor an adaptation procedure; it measures transfer on the declared source/domain strata.

2.4 Calibration at Low False-Positive Rates

Axelsson's (2000) base-rate analysis explains why rare threats can produce alert populations dominated by false alarms. It motivates specificity, not this study's particular 1% ceiling. Validation therefore maximizes recall subject to a one-sided 95% Clopper–Pearson FPR upper bound <=1% (Clopper & Pearson, 1934), with fixed ties and no test-informed relaxation. Binomial bounds require an appropriate independence model; repeated-domain uncertainty is addressed separately in Section 3.8.

ROC AUC describes ranking across thresholds, whereas precision–recall views expose behavior under class imbalance (Saito & Rehmsmeier, 2015). MCC summarizes all confusion-matrix categories but does not encode a gateway's risk budget (Chicco & Jurman, 2020). Proper scoring rules evaluate probability assignments rather than prescribe actions (Gneiting & Raftery, 2007). These complementary summaries accompany, rather than replace, the locked operating-point counts.

Prevalence projections hold measured class-conditional rates fixed while varying an assumed class mixture. This resembles the label-shift distinction studied by Lipton et al. (2018), but estimates neither deployment prevalence nor that paper's correction method. Improved ranking, calibrated probabilities and acceptable alert burden remain separate empirical questions.

2.5 Selective Cascades

A selective cascade applies a less expensive model first and invokes a more expensive model for a defined subset of inputs. CascadeBERT demonstrates calibrated language-model cascades as a way to exchange computational cost for prediction quality (Li et al., 2021). Alajaji (2026) evaluates a validation-selected classical-first cascade with transformer deferral for phishing email-body detection; that result establishes modality-specific deferral prior art, not raw-URL or inference-gateway performance. Hussain et al. (2027) combine calibrated URL experts through confidence- and uncertainty-aware routing and evaluate cross-dataset shift with registered-domain-stratified splits. Their routing combines expert decisions rather than implementing a cheap-first deferral policy, so it does not establish transformer-call savings for the present cascade. Together, these studies establish model deferral and calibrated expert combinations as prior art while leaving the present operating claim to be tested.

The fixed cascade in the present study uses a symmetric uncertainty band around the locked Logistic-L1 threshold. Validation candidates consist of unique absolute distances from that threshold. The selected band is the one with the fewest escalations whose validation cascade recall is at least transformer recall minus 0.02 and whose FPR upper bound is <= 1%. Test data do not select or revise the band.

Confidence-based deferral is attractive only if the selected cases are ones for which additional computation helps. Ovadia et al. (2019) evaluate predictive uncertainty under dataset shift and find that conventional post-hoc calibration can fall short outside the original distribution. Their benchmark is not a URL cascade, and the present system does not implement the ensemble methods they compare. The relevant implication is that

low confidence, high error probability and useful escalation cannot be treated as synonyms under shift. A confidently wrong first-stage decision may fall outside an uncertainty band; a deferred case may remain wrong after the second stage. The present evaluation therefore accounts separately for transformer invocations, final reference-label errors and service cost. A reduction in calls has value only in relation to the detection behavior retained at that call rate.

2.6 Distribution Monitoring and Routing

A GMM models reference features through weighted Gaussian components, fitted by expectation–maximization (Dempster et al., 1977). The present monitor selects among one through six diagonal components using BIC (Schwarz, 1978), scores windows by mean negative log-likelihood and separates threshold calibration from false-alert auditing.

CADE detects and explains drift samples in malware and intrusion settings (Yang et al., 2021); it does not validate this monitor's particular likelihood or routing rule. Gama et al. (2014) and Lu et al. (2019) distinguish detecting change from understanding and adapting to it. A high likelihood departure concerns P(X), not direct evidence that P(Y| X), labels or prediction errors changed. Escalating future requests is a separate action requiring an error comparison.

Anomaly scores are relative to the fitted reference (Chandola et al., 2009). Benign records can be unusual and harmful records familiar. MMD supplies a kernel-based distribution comparison (Gretton et al., 2012), not a guarantee of phishing-error improvement. Accordingly, the evaluation retains distinct denominators for window

detection, reference alerts, request errors and added model execution. Overlapping windows and future-only activations also limit independence-based inference.

2.7 Gap Addressed by the Study

The reviewed studies establish URL representations, calibrated deferral, cross-source analysis and distribution monitoring as substantive research areas. Their evidence differs, however, in the inputs available to the detector, the populations on which it is evaluated and the costs included in measurement. The problem addressed here is their joint operation at a gateway boundary: whether a compact two-stage URL-only detector can retain low-FPR recall across internal and external evaluation, limit transformer invocation and satisfy an end-to-end HTTP budget under fixed procedures. This question requires paired detection outcomes, explicit routing traces and service measurements for the same implemented system. Table 2.1 identifies how the closest studies inform that problem and where their evidence does not substitute for its evaluation.

Table 2.1. Closest prior work and the boundary of the present contribution.

Closest work	Setting and contribution	Boundary relative to this study
Ahamed et al. (2026)	Raw-URL robustness across structural, character, adversarial, domain-disjoint, and external checks	Includes separate PhishTank/Majestic evaluation and normalization/robustness interventions. The present comparison adds same-example expert repair, a fixed risk constraint and measured invocation/service costs; cross-source collapse itself is not a new discovery.
Rashid et al. (2024)	Cross-dataset phishing-URL generalization through unsupervised domain adaptation	Adapts across sources, whereas this study freezes the detector and prohibits external refitting or recalibration.
Tsai et al. (2024)	Diagnosis and adversarial mitigation of dataset bias in malicious-URL models	Studies invariant representation learning rather than a low-FPR cascade, monitor, or service evaluation.
Li et al. (2021)	Calibrated complete-model cascades for general language-model inference	Establishes cascade prior art outside phishing and does not test the present URL or gateway constraints.
Alajaji (2026)	Validation-selected classical-first transformer deferral for phishing email bodies	Uses a different modality and does not establish raw-URL or measured inference-gateway behavior.

Closest work	Setting and contribution	Boundary relative to this study
Hussain et al. (2027)	Calibrated URL-expert fusion under cross-dataset and target-prior shift	The author repository describes prior-specific experts and calibrated routers using unlabeled target statistics. This study tests conditional repair by a selected gate and its actual cost; it claims neither priority for expert routing nor a full-paper replication.
Yang et al. (2021)	Detection and explanation of individual security drift samples	Does not establish the present GMM, window construction, false-alert audit, or routing rule.

The contribution therefore lies in the evaluated relationships among these components. A representation can improve recall without transferring its false-positive behavior; an alert can recognize source departure without selecting requests that benefit from escalation; and avoided inference calls need not produce acceptable HTTP latency. The following sections establish the methodological basis for examining those relationships and for testing specific modifications after the initial evaluation.

2.8 Evaluation-Guided Engineering Iteration

Arp et al. (2022) describe how spurious correlations, inappropriate evaluation sets and test-informed development distort security-ML conclusions. A diagnostic observation therefore motivates a mechanism test; it does not establish a cause. TESSERACT's spatial/temporal distinction likewise warns that domain exclusion does

not establish forward-in-time performance (Pendlebury et al., 2019). The additional 2020 benchmark predates the reported PhiUSIIL phishing collection and shares upstream feeds, making D retrospective.

Guo et al. (2017) calibrate confidence using validation data. In binary prediction, a positive-temperature monotone transform cannot improve ROC ranking, so recalibration cannot automatically repair discrimination. SelectiveNet evaluates risk with an explicit coverage/rejection constraint (Geifman & El-Yaniv, 2019); deferred cases are not necessarily corrected predictions. Neither algorithm is implemented here.

Chen et al. (2018) describe metamorphic relations for testing specified behavior across transformed inputs. D's opposite-scheme companions test exact feature and score invariance, not equivalence of live HTTP/HTTPS pages. The broader iteration separately measures representation conformance, labeled detection and service performance. It connects an observed defect to a bounded modification and a comparison that can reveal both improvement and cost.

2.9 Operational Measurement and Evidence Units

Connection ownership, scheduling, queueing and response validation contribute to an endpoint's latency. Dean and Barroso (2013) motivate examining tails rather than averages. Kalibera and Jones (2013) emphasize replication at the experimental levels that vary. S therefore uses paired runs for ratio uncertainty and reports pooled request quantiles separately; a large request count is not the same as many independent experiments.

Closed-loop clients submit after prior completion, unlike externally timed open arrivals (Schroeder et al., 2006). The measurements characterize that closed-loop path,

not unrestricted production capacity. Predictive-field equality also differs from exact response equality: concurrency can change admission sequence without changing classification, and timeouts create noncomparable pairs. Full attempts, errors and both agreement definitions remain visible. These distinctions connect detection quality, routing, computation and HTTP cost without allowing one favorable metric to stand in for the entire system.

Chapter 3—Methodology

3.1 Research Design

The staged-freeze design evaluates representation (RQ1), monitoring and future-only routing (RQ2), and real-HTTP execution (RQ3). Detection outcomes are retained URL records; recall resampling clusters them by registrable domain. Monitoring uses complete 256-request windows at stride 64. Operational errors use measured HTTP requests. The original 125-cell matrix has two accepted sessions: a complete retained prefix of 72 cells and a continuation of 53. Models, thresholds, workloads and decision rules remain fixed. Section 4.6 and Appendix C explain the recovery chronology; no favorable subset was selected.

3.1.1 Architecture and Information Flow

Preparation establishes eligibility and identity before inference. Each URL yields 25 structural features and a Logistic-L1 score; the fixed uncertainty band either retains that decision or invokes the character transformer. The GMM observes structural features and the first-stage score, with alerts affecting only future requests. Evaluation joins saved outcomes to publisher-label strata and measures HTTP execution separately (Figure 3.1; Table 3.1).

After mechanical quarantine, every retained external record enters the same ordered, label-blind routing stream. Gold, certified and secondary designations define analysis strata afterward; they do not select a source-aware policy. Logical routing and physical model execution are recorded separately.

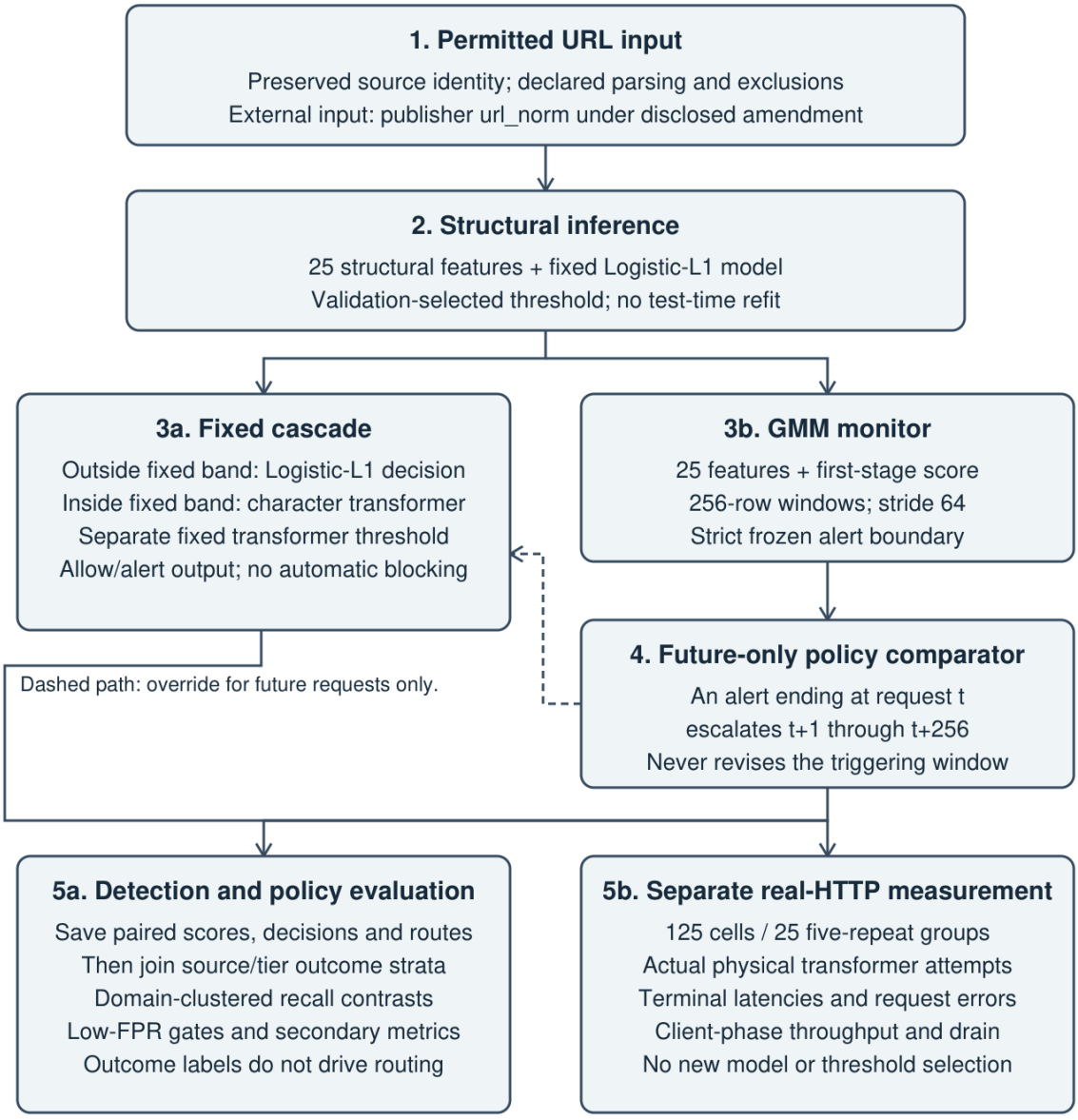


Figure 3.1. Frozen inference, future-only routing and separate evaluation boundaries.

Table 3.1. Reading map from technical component to experiment and evidence.

Plane	Technical operation	What its evidence establishes
Preparation	Preserve source fields; apply declared parsing, quarantine, domain allocation and overlap checks	Which records and domains enter each denominator; not independently adjudicated labels
Representation	Length-only, 25-feature Logistic-L1, character transformer and fixed cascade	Paired recall differences and specificity at unchanged thresholds; RQ1, Tables 4.2–4.3
Monitoring and policy	Training-fixed GMM; validation-calibrated windows; next-256 escalation	Distribution sensitivity, reference false alerts and the policy's separate error consequences; RQ2, Tables 4.4–4.5
Service	Real HTTP client/service processes; physical inference attempts; terminal outcomes and latencies	End-to-end cost, failures and throughput for every scheduled repeat; RQ3, Tables 4.6–4.7
Verification	Bound artifacts, retained-cell eligibility, process exits and saved-evidence recomputation	Traceability from accepted execution to all 22 decisions; not independent replication or institutional approval

3.2 Outcome-Label Contract

The local is_phishing field denotes the publisher reference classification: 1 means phishing and 0 legitimate. PhiUSIIL native label 0 maps to 1, and native label 1 maps to 0. External mappings are verified categorically rather than assumed numeric: NCSC gold designations supply primary positives and certified trusted-registry designations primary negatives. NCSC silver and ChongLuaDao/OpenPhish bronze retain secondary roles.

Tranco is a reference-negative control, not verified benign ground truth. These labels are not new forensic judgments.

External eligibility requires a valid URL, source, source class, confidence tier, published split and stable identifier. Unknown mappings are quarantined. Conflicting canonical-URL groups, domains crossing development or published external splits, and external domains overlapping PhiUSIIL are excluded as whole groups. Same-label duplicates retain the lexicographically smallest stable identifier; groups without one are quarantined. No personal adjudication overrides these mechanical rules. Outcome strata are applied only after the retained stream has been routed.

3.3 Development Source and Preservation

PhiUSIIL, UCI dataset 967 (CC BY 4.0), is the sole fitting, validation and internal-evaluation source (Prasad & Chandra, 2024b). Its archive and CSV identities are retained in the method supplement identified by Appendix C. The publisher identifies Open PageRank as the legitimate source and PhishTank, OpenPhish and MalwareWorld as phishing sources, with phishing retrieval from October 1, 2022 through May 21, 2023 (Prasad & Chandra, 2024a). A corresponding legitimate collection window and independent per-row collection-source, timestamp and adjudication fields are not established. These metadata boundaries constrain source-specific and temporal interpretations.

3.4 Canonicalization and Quarantine

Preparation accepts absolute HTTP(S) URLs. It lowercases the scheme, IDNA-normalizes the hostname, normalizes numeric ports and removes defaults, represents an empty path as '/', and uppercases percent-escape hex without decoding. User information,

path spelling, query order and fragments remain. Canonical keys identify duplicates and conflicts; labels never derive from those keys. Unsupported inputs and conflicting groups receive mechanical exclusion reasons.

Original external preparation used publisher url cells and reached a whole-study hold. The disclosed amendment instead uses exact publisher url_norm for both parsing and model input, retaining raw cells as provenance. It permits no local scheme repair, trimming or raw fallback. Mapping, overlap and quarantine rules remain unchanged. Adoption followed preparation and label-count inspection, before amended predictions; Section 4.6 retains that information boundary.

3.5 Registrable-Domain Allocation

A pinned Public Suffix List defines registrable domains. Domain groups are ranked by SHA256 of the allocation identifier, seed 20260816 and domain. Hamilton largest-remainder allocation targets 70% train, 15% validation and 15% group test, with ties resolved train, validation, test. Allocation is label-blind, without stratification or rerolls. Training fits preprocessing and models; validation selects thresholds, checkpoints, cascade bands and monitor boundaries. Group test never enters fitting or selection.

Text searches displayed group-test rows after the baseline freeze but before scoring, establishing analyst exposure. They prompted no model or threshold change. Later interrupted attempts repeated source scoring under unchanged rules. Consequently, the realized evaluation is neither an unseen test nor a single lifetime pass. Appendix C and the full method supplement preserve the dates and sequence; recovery does not restore blindness.

3.6 Detection Models

The structural extractor produces an ordered 25-element float64 vector from the supplied string, urlsplit components and IDNA-normalized hostname, without destination access. Features describe six lengths; six hostname, path and query counts; eight character-composition, entropy and escaping quantities; and five scheme/component indicators. Their exact names, order and parsing rules remain in the complete method supplement. Correlated features make individual coefficients unsuitable as causal importance estimates. Missing or invalid raw_url raises an error rather than imputation. Labels, split and record identities, canonical/domain keys and publisher provenance fields are forbidden predictors.

Length-only and Logistic-L1 use training-fitted StandardScaler followed by LogisticRegression: SAGA, L1, C=1, balanced class weights, an unpenalized intercept, max_iter=5000, tol=1e-4 and random_state=42. The former uses raw codepoint length; the latter all 25 features. Convergence must occur before the iteration limit, with finite fitted state. There is no hyperparameter search. The accepted SAGA design followed a failed liblinear fit and training-only diagnostic before accepted validation. Its historical warnings and corrections remain separately documented.

The scoring-integrity rule narrowly permits three recorded macOS/Accelerate RuntimeWarnings during decision_function or predict_proba, only when all outputs are finite and both decision values and the full probability matrix match independent float64 arithmetic at rtol=atol=1e-12. Other stages and warnings remain fatal. Accepted evaluation preserves the authoritative scoring procedure, including floating-point ties.

Stage 2 canonicalizes the URL, percent-encodes remaining non-ASCII characters as UTF-8, and preserves existing escapes and RFC 3986 reserved characters (Berners-Lee et al., 2005). Sequences longer than 256 characters retain the first 192 and last 64. The training-only ASCII vocabulary reserves PAD=0 and UNK=1; sequences are right-padded with PAD masked. The float32 PyTorch 2.7.1 transformer has 192-wide token and position embeddings, four pre-normalization layers, six heads, feed-forward width 768, GELU and dropout 0.1. Final LayerNorm and masked mean pooling feed a linear logit head.

The MPS training runtime disables mixed precision, enables strict deterministic algorithms, uses zero loader workers and seed 42. Binary cross-entropy uses pos_weight equal to the training negative/positive ratio. AdamW uses learning rate 1e-4, betas (0.9,0.999), epsilon 1e-8, weight decay 0.01 and no scheduler (Loshchilov & Hutter, 2019). Training/validation batches are 256/512; gradient clipping is 1.0 and the epoch cap 40. One seeded generator advances across shuffled training epochs; ordered validation retains final partial batches.

Validation AP is computed after every epoch with dropout disabled. Epoch 1 establishes the best; improvement must exceed min_delta=1e-4. Five consecutive nonqualifying epochs stop training, and the earliest qualifying best is restored. Warnings, nonfinite values, missing classes or deterministic-operation errors halt publication. Stage 1 remains the accepted, unrefitted Logistic-L1 artifact. The full method supplement retains initialization constants, exact serialization and publication rules, failed predecessors and input/output identities. These bind fitted state to predictions; they do not establish external efficacy.

3.7 Threshold and Cascade Calibration

An alert occurs at score >= threshold; the score is not assumed to be calibrated deployment risk. Unique validation scores plus nextafter(max_score,+infinity) form the candidates. Selection maximizes recall subject to the one-sided exact 95% Clopper–Pearson FPR upper bound <=1%, breaking ties by smaller bound then higher threshold. Infeasibility records target_not_met without relaxing the constraint.

Both component thresholds must be selected before cascade calibration. Inclusive symmetric bands satisfy abs(stage1_score-stage1_threshold) <= half_width, with candidate widths given by unique validation distances. Inside the band, transformer score and decision replace stage 1; outside, Logistic-L1 remains. Feasible bands require recall >= transformer recall minus 0.02 and the same FPR-bound constraint. Selection minimizes calls, then width. The selected thresholds and band remain unchanged during evaluation and replay.

3.8 RQ1 Analysis

The September 21 inference-method amendment adopted the previously specified singleton OpenBLAS/MPS convention for subsequent no-fit evaluation and serving. The development comparison did not establish exact equivalence with the original batch-scoring convention: one transformer decision differed. Adoption was therefore a development-informed methodological change before protected evaluation, not a passed compatibility test. Historically selected weights, thresholds and the cascade band were carried forward without recalibration. Their optimality or checkpoint-selection equivalence under singleton scoring is not claimed. The accepted paired predictions use the common convention; historical development summaries retain their original meaning.

The amendment did not authorize an alternative-runtime search or additional compatibility execution.

Length-only, Logistic-L1, transformer-only, and cascade predictions are paired by retained record. The original RQ1 plan specified one noninteractive group-test pass and one external pass after their respective freezes. Source scoring subsequently occurred in repeated interrupted study attempts; the first, second and fourth amended attempts accepted both source outputs. Each attempted source evaluation used the same frozen models and thresholds, with external source and confidence filters applied after routing to form the outcome strata. Final synthesis uses the authenticated evidence selected under the disclosed recovery protocol, without choosing among attempts by their outcomes. Repeated execution does not restore test-set blindness or establish compliance with the original single-pass intention.

Bootstrap resampling follows the general framework of Efron (1979), with the domain-clustered estimand specified below. The original primary recall analysis used a registrable-domain-clustered percentile bootstrap using 2,000 replicates and seed 20260816. H1 uses the lower confidence bounds for the four prespecified differences and the observed FPR gates in Section 1.4.1. Average precision (AP), precision, F2, Matthews correlation coefficient (MCC), balanced accuracy, receiver operating characteristic area under the curve (ROC-AUC), Brier score, and calibration error are secondary descriptions.

The paired-recall implementation operates on saved binary predictions for the declared positive stratum. Record identifiers must be unique and match in the same order across both prediction vectors. Let D be the number of registrable domains represented in

that stratum. Domains are ordered by canonical ASCII spelling, and each replicate samples D domains uniformly with replacement. Every occurrence of a selected domain contributes all its eligible positive rows. The replicate statistic is the sum of candidate-minus-reference decisions divided by the sampled row count. This preserves URL weighting when domains have unequal sizes; averaging domain-level recall would estimate a different quantity.

Each contrast uses a fresh NumPy 2.2.6 Generator(PCG64(20260816)), with one integers(0, D, size=D, dtype=int64) call for each of 2,000 replicates. Contrasts on the same positive stratum therefore use the same domain draws. The 2.5th and 97.5th percentiles use linear interpolation. An empty stratum has no estimate or interval. A stratum containing one domain retains its point estimate but has no estimable clustered interval. With at least two domains, the historical implementation returned any degenerate interval as calculated. The corrected decision analysis below does not use such an interval as evidence of population certainty or noninferiority. The procedure is implemented and checked on synthetic known-answer fixtures in paired_evaluation.py. Runtime, manifest, operational and secondary-analysis specifications were subsequently bound to the execution profile used for the attempts in Section 4.6; Chapter 4 reports the completed, verified evidence under the disclosed recovery boundary.

Post-review paired-inference correction. Identical observed predictions make every bootstrap replicate identical; resampling cannot introduce an error pattern absent from the sample. A zero-width interval therefore cannot establish the two-percentage-point population recall margin. The original outputs remain in the research record, but the corrected adjudication replaces that interpretation. The original research questions,

outcome strata, numerical margins, observed-rate gates and two-sided 95% confidence convention are retained. This is an analysis correction after review, not a claim that the replacement method was prespecified before the study.

For nonstateful gold contrasts, let b denote candidate-only detections and c reference-only detections among n labeled positive pairs. The estimand is the difference between their detection probabilities, estimated by (b-c)/n. Each discordant count has a binomial marginal when pairs are independent. Clopper–Pearson bounds allocate 0.05/4 to each of the four marginal tails; projecting their simultaneous rectangle gives [L(b)-U(c), U(b)-L(c)]. The union bound supplies at least 95% coverage under independent, identically distributed paired sampling, without assuming independence between b and c. This conservative envelope is not an optimized shortest exact interval. Newcombe's (1998) method 10 is reported separately as an approximate-score sensitivity analysis, not as the source of the exact guarantee. For 69 concordant positive pairs, the selected envelope is [-6.1533, 6.1533] percentage points; method 10 gives [-5.2737, 5.2737]. Neither establishes the -2-point lower-bound requirement. Failure to establish that margin does not demonstrate that the cascade is inferior.

The 69 gold rows belong to distinct registrable domains, but domain uniqueness alone does not establish campaign independence or label accuracy. Those are assumptions of the gold-pair intervals. The nondegenerate internal structural-versus-length contrast retains the original whole-domain bootstrap and URL-weighted estimand. An independent-row formula is not substituted for repeated-domain internal data or for a policy whose predictions depend on preceding stream windows. Their degenerate zero contrasts retain their observed estimates without an established population lower bound.

Likewise, historical row-wise binomial FPR bounds are not population-coverage guarantees for the 2,497 certified negatives in 234 domains. Original FPR decisions continue to use observed rates, as specified. These are contrast-specific intervals, not simultaneous 95% coverage for all 22 checks; the joint hypotheses still require every constituent condition.

3.9 GMM Monitoring and Future-Only Routing

The monitor standardizes the 25 structural features plus the accepted Logistic-L1 probability using training-only population variance (ddof=0; constant-column scale=1). Its separate portable stage-one path remains unchanged. CPU float64 scoring uses one library thread and pinned NumPy 2.2.6, SciPy 1.15.3 and scikit-learn 1.7.2 with the required OpenBLAS backend. Validation domains are hash-ranked with seed 20260816; the first floor(D/2) form calibration and the remainder a disjoint false-alert audit, preserving source order within each stream.

Six training-only diagonal GMMs use 1–6 components, five k-means starts, random_state=42, tolerance .001, covariance regularization .000001 and max_iter=500. Lowest training BIC selects the model, breaking ties toward fewer components (Schwarz, 1978). All candidates must converge with finite, valid parameters; failures are not skipped or retried. Numerical underflow is permitted only for mixture fitting, BIC and likelihood calculations. The full supplement retains the exact numerical-error policy.

Complete 256-request windows at stride 64 are scored by mean negative log-likelihood. The alert threshold is the calibration scores' linearly interpolated 95th percentile, with strict greater-than comparison. The audit passes only when 20×alerts <= complete windows; its boundary is not retuned. Every complete external window is a

prespecified shift window, including overlapping windows as separate observations. An incomplete terminal window does not enter the rate. Neither fraction receives an independent-binomial interval.

External file order is preserved. The first 256 requests use the fixed cascade; an alert at each eligible window end affects exactly the next 256 requests. Activations are unioned and truncated at stream end, with at most one transformer invocation per request. This is future-only offline replay, not a randomized field intervention. Saved paired policy-minus-cascade recall equals false-negative reduction; resampling those outcomes does not rerun routing and remains conditional on the observed order. Shared activations create dependence beyond domains. The monitor detects P(X) departure, not necessarily harmful P(Y|X) change. MMD, PSI and string probes remain secondary diagnostics.

3.10 External Evaluation Freeze

External evaluation uses the complete retained PhishVN published test split in file order, with categorical strata applied after label-blind routing and no model refit or recalibration (Vu, 2026b). Mendeley Version 4 distributes PhishVN v3.1.0; its metadata describes a documentation-only update from Version 3 with byte-identical data files and added label-audit documentation.

The initial freeze bound source rules, prepared populations, fitted artifacts, thresholds, gate, monitor, evaluator, manifest and runtime. Input adoption later followed preparation and label-count inspection; bounded recovery followed source predictions and partial measurement. Thus later amendments cannot inherit the initial prediction-blind description. Section 4.6 and Appendix C distinguish scientific rules, information exposure and execution repair. Operator authorization is not advisor approval.

3.11 RQ2 Analysis

H2 uses the four unchanged gates in Section 1.4.2: complete-window external sensitivity, disjoint validation-audit false alerts, certified-negative policy FPR and positive paired recall improvement. The original 2,000-replicate whole-domain bootstrap describes saved policy outcomes conditional on the stream; Section 3.8 corrects unsupported uncertainty interpretations. These comparisons evaluate the action following the monitor signal, not a causal effect attributable solely to that signal.

3.12 RQ3 Operational Replay

The primary manifest samples 10,000 group-test URLs without replacement at a designed 1% positive prevalence; 0.1% and 5% are sensitivities, not production estimates. Seed 20260816 fixes class sampling and final order with separate PCG64 purpose streams. Candidates sort by stable source-row ID, every repeat uses the same order, and the first 1,000 rows provide excluded warmup. Insufficient class yield cannot trigger replacement sampling or a changed target. Manifest identities bind exact raw spelling.

The 125-cell schedule comprises 25 five-repeat groups: 90 fixed-cascade cells at three prevalences and concurrencies 1,8,16,32,64,128; 30 transformer-only 1% cells; and five serialized live-shift cells. Ordinary arms have 1,000 warmups and 10,000 measured requests; live-shift arms measure 8,701 external rows after separate warmup/reset. H3 uses all 50,000 measured fixed-cascade 1% concurrency-64 requests. Pooled p95 includes failures and uses the linear quantile, not averaged run quantiles.

Each run starts a fresh service/client with one inference owner and a FIFO capacity of 128 waiting requests. Closed-loop workers issue the next request after

termination of the preceding one (Schroeder et al., 2006). Timing covers submission, body read and response validation, including connection, queue and processing, but excludes unsent backlog. Success requires HTTP 200 and valid identity-matched JSON within 2,000 ms. No request is retried; actual terminal latency is not clamped. Timeouts, malformed responses, transport failures and other statuses remain errors in the full denominator. Admitted work continues after client timeout; phase closure prevents late admission and drains work before counters are recorded.

Physical invocation evidence uses the prospectively designated first concurrency-1 primary run. H3 retains the external safeguards, two-point recall margin, <=30% invocation, <=200-ms p95 and <0.1% error limits in Section 1.4.3. The accepted two-session matrix retains all 125 ordinals: cells 1–72 from one admitted prefix, then 73–125 measured in order with cell 73 restarted in full. Primary cells 1 and 21–25 remain within the first session; descriptive group 71–75 crosses the boundary. Session and workload order are confounded. The complete schedule and errors remain in the research record.

3.13 Reproducibility and Integrity Controls

Versioned contracts identify inputs, fitted state, thresholds, population rules and denominators. Independent saved-observation calculations reconstruct counts, paired outcomes and latency summaries; they are arithmetic verification, not independent empirical replication. Appendix A maps evidence to code; Appendix C and the full method supplement preserve histories and hashes. The GMM's failed development false-alert gate and all adverse model outcomes remain part of the results, with no relaxed primary requirement.

Secondary analyses include formatting-only and Random Forest comparators, five historical shuffled-label fits, seed/runtime sensitivity, MMD/PSI, label-free probes, calibration and paired tests. Score summaries include AP, ROC AUC, empirical recall at FPR <=1% without interpolation, Brier score and ten-bin ECE. Operating-point decisions are separate from scores because cascade stages have different thresholds. Four exact McNemar contrasts compare structural versus length and cascade versus structural in the two positive populations, retaining all slots under Holm adjustment (McNemar, 1947; Holm, 1979). These nominal paired tests do not model domain dependence or replace primary intervals.

A source-only classifier requires per-row provenance independent of outcome; PhiUSIIL lacks it, so source cannot be imputed from class. Formatting uses five indicators. Historical label permutations use fresh PCG64 seeds 42–46, fixed Logistic-L1 solver seed 42 and unchanged validation labels, with separate CP cutoffs. The Random Forest retains its accepted direct-leaf arithmetic. MMD/PSI references fit training data before validation and retain the monitor's calibration/audit partition with separate strict 95th-percentile boundaries. No comparator replaces a primary artifact.

Character seeds 43–46 reuse the training-only vocabulary and original 256/512 batching and stopping rule; seed 42 is not refitted. Every checkpoint and validation vector is retained. All five sets use the common singleton secondary scorer and individually selected secondary thresholds/bands. Seed 42's earlier training runtime makes this seed/runtime sensitivity, not isolated seed causality. Probe streams preserve every position, including no-ops and ineligible strings; transformed strings receive no inherited correctness labels. Models, boundaries and scalers stay fixed, and monitor history resets

per stream. Section 3.16 adds direct fit-boundary evidence for new shuffled-label controls; it does not authenticate historical fits.

3.14 Ethical and Safety Considerations

The study treats URLs as inert strings and does not contact their destinations. The original versioned research release distinguishes licensed source records and model evidence from private execution files. For the post-review cohort, URL strings, domains, row identifiers and record-level predictions remain private; the public supplement contains aggregate results, specifications, code and permitted file-level identities. This narrower release boundary preserves verification without exposing newly retained identities. The gateway returns alerts for review rather than performing automatic blocking or takedown.

Manual review is permitted only as separately reported post hoc descriptive error analysis. It cannot assign or override labels or change quarantine, inclusion, thresholds, features, models, gates, or hypothesis decisions.

3.15 Bounded Follow-up Design and Evaluation

3.15.1 Sequence and Scope

D and S were specified on October 1, after the original findings but before additional benchmark retrieval, candidate fitting or final client measurements. D tests consistent scheme neutralization; S tests worker-owned persistent HTTP clients. Neither changes the transformer, GMM, H1–H3 or their requirements. Each primary comparison uses a two-sided 97.5% bootstrap interval, allocating the error budget across the two comparisons without an independence assumption; finite-sample coverage remains

approximate. The extended method supplement preserves the chronology and authorization boundaries.

3.15.2 Additional Benchmark and Eligibility

The CC BY 4.0 Web page phishing detection dataset, Mendeley Version 3, contains 11,430 URLs with equally sized publisher-labeled classes (Hannousse & Yahiouche, 2020, 2021a, 2021b). Only URL and class are used, not its 87 features, stored page objects or live destinations. Metadata and the inspected preprint disagree on the collection month, so it is identified as the 2020 benchmark. Alexa/Yandex legitimate-source mechanisms differ partly from development, while PhishTank/OpenPhish feeds overlap. This is retrospective reference-label evidence, not an independent production sample.

Before scoring, original canonicalization and suffix rules exclude invalid URLs, mixed-label canonical duplicates and domains in full retained PhiUSIIL/PhishVN manifests. Both raw and normalized PhishVN fields contribute parseable domains; 50,832 unparseable field instances do not equal missing records. Same-label duplicates retain first source order. The whole-study admission minimum is 1,000 rows and 250 domains per class, not a power guarantee.

All 8,622 eligible rows are included: 4,651 positives in 2,897 domains and 3,971 negatives in 3,377 domains. One domain spans classes, giving 6,273 total domains. No outcome-based sampling occurs. Figure 3.2 separates development, diagnosis, overlap screening and evaluation: original training/validation fit and calibrate the candidate; the additional benchmark selects neither it nor its operating point.

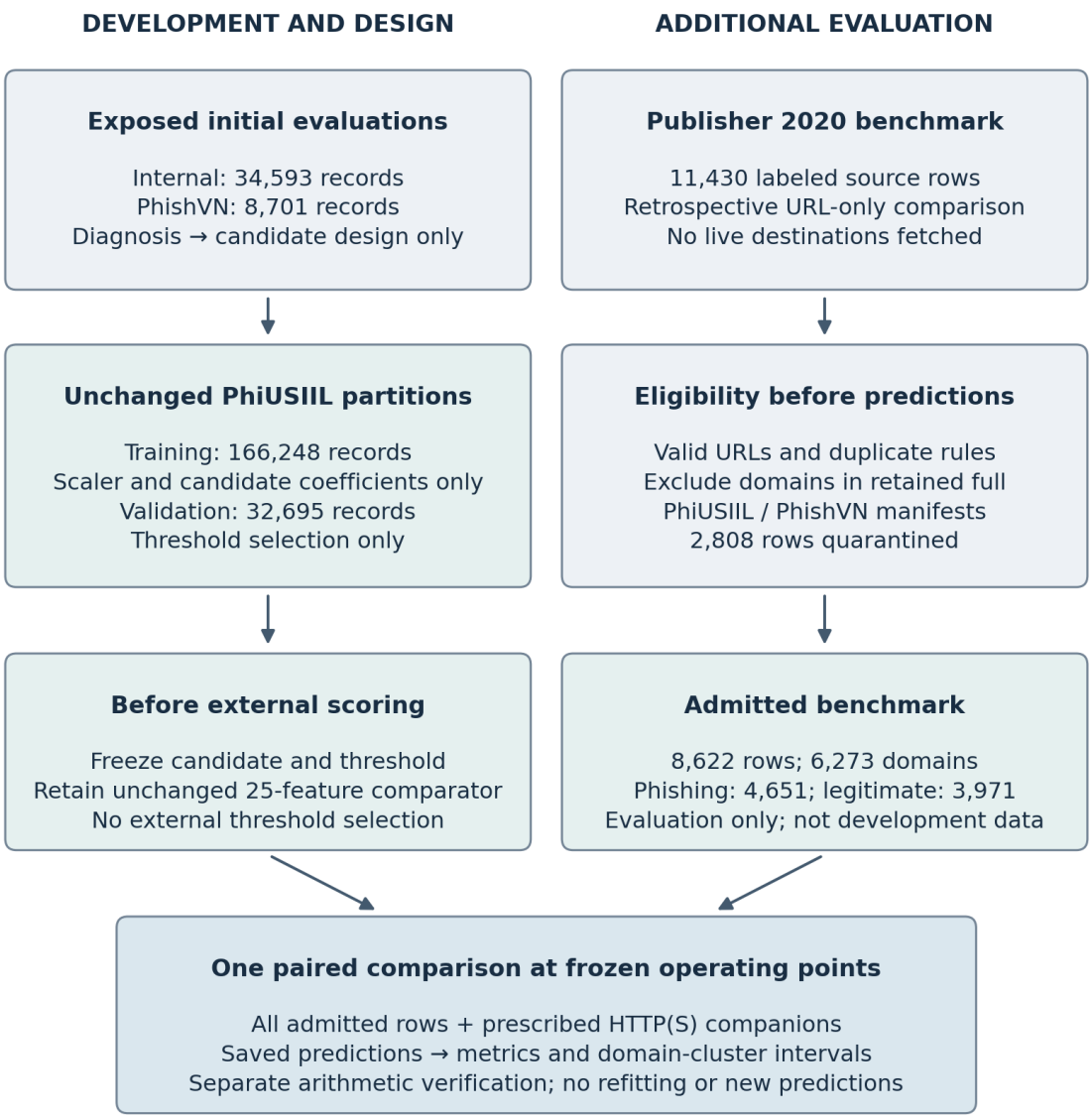


Figure 3.2. Source and partition roles in the bounded detection comparison. Arrows show permitted information flow; domain exclusion does not establish temporal or source-mechanism independence.

3.15.3 Transport-Neutral Structural Representation

The accepted 25-feature comparator remains unrefitted at threshold 0.2670846328466124. The candidate validates absolute HTTP(S), replaces only the scheme spelling with http and derives the original structural features except is_https. This

24-feature contract removes scheme effects from lengths, counts, ratios and entropy as well as the indicator, preserving raw evidence.

A single fit uses the original 166,248 training rows and training-only scaling, with the Logistic-L1 configuration in Section 3.6, one thread and seed 42. The unchanged maximum-recall/FPR-bound rule uses only the original 32,695 validation rows. The fit converges in 4,971 iterations and selects threshold 0.6815549262616749: validation TP=7,588/12,486 and FP=178/20,209, giving recall 60.77%, FPR 0.8808% and upper bound 0.9968%. These select the operating point, not external efficacy.

Every admitted URL has one opposite-scheme companion. Exact feature, score and decision equality is required, using singleton sklearn scoring and the independent float64 audit. These are metamorphic representation checks, not newly labeled websites or claims of live-page equivalence. Refitting and recalibration make D a combined intervention rather than a single-feature causal ablation.

3.15.4 Detection Estimands and Uncertainty

D requires candidate-minus-comparator recall gain >=5 percentage points, a positive lower endpoint of its two-sided 97.5% interval, observed candidate FPR <=1% and exact scheme invariance. The comparison uses original eligible strings. Ten thousand paired PCG64 domain resamples, seed 20261001, select from 6,273 sorted domains; each selected domain contributes all its rows with multiplicity. Recall and FPR remain row-weighted. Undefined class denominators are counted and excluded only from the corresponding interval. Confusion matrices, precision, ROC AUC, AP, Brier score and ten equal-width calibration bins remain secondary; undefined values are not replaced with zeros.

3.15.5 Service Comparison and Its Boundary

S compares a shared persistent client pool with one persistent client per closed-loop worker, holding endpoint, input order, response contract, two-second deadline and no-retry rule fixed. Its no-model and singleton structural workloads contain 10,000 synthetic .example URL occurrences, without labels or asserted prevalence. Neither invokes the transformer. The no-model control separates classifier work from the broader client/service path, not each internal scheduling cost.

Concurrency 1 and 64, ten pairs per workload/concurrency combination and two topologies give 80 fresh-service arms. Odd/even pairs reverse topology order. Every arm has 1,000 warmups, 10,000 measured attempts and phase drains. A preceding interrupted schedule contributes no primary estimate; the replacement rule preceded any primary structural concurrency-64 arm. The complete chronology remains in Appendix C.

Primary S is the median of ten paired worker/shared run-p95 ratios for structural concurrency 64. Ten thousand paired PCG64 resamples, seed 20261002, produce the two-sided 97.5% interval. Requirements are ratio <=0.8, upper endpoint <1, pooled successful worker p95 <=200 ms, errors <0.1% of all 100,000 primary worker requests and exact paired responses excluding request IDs only. Admission sequence remains in that last requirement; prediction-field agreement is reported separately. Ten pairs limit bootstrap resolution.

Sampled host observations and separate success, error, latency, throughput, physical-counter and drain records describe the measured conditions. They do not establish a continuously disturbance-free host. S measures synthetic structural-only

closed-loop HTTP execution, not transformer acceleration, production capacity or a replacement for original H3.

3.16 Post-review Diagnostic and Routing Comparisons

The October 12 post-review study addresses the mechanisms raised by the original evaluation without reopening model selection. Its specification fixes four linked tasks before any new fit or cohort prediction: F, a frozen-model formatting intervention; R, a domain-balanced retrospective routing comparison; C, actual routed-computation measurement; and P, five audited shuffled-label fits. Every task requires eligible inputs before the study begins. Original H1–H3, D and S remain separate, and no new support for H2 is inferred from these tasks.

3.16.1 Frozen-model Intervention and Domain-balanced Sampling

F uses all 8,701 previously evaluated external strings. It preserves labels, the accepted structural weights and scaler, and threshold 0.2670846328466124. Each retained spelling is compared with forced http: and https: prefixes, preserving every character after the first colon. Each form is scored once; the original form must reproduce its retained result. Paired features, scores and decisions identify the spelling effect on the frozen model. Changing a string is not an intervention on a live website and does not isolate a single feature, because the prefix enters several structural features.

R uses retained PhishVN publisher train/validation records (Vu, 2026a, 2026b), not the publisher test partition. Positive eligibility requires the tinnhiemmang source, gold tier and phishing label. Negative eligibility requires the tinnhiem_org or tinnhiem_web source, gold tier and benign label; label_source must be present. The unchanged canonicalizer rejects invalid URLs, contradictory canonical labels and hostname

disagreement. Domain exclusions cover publisher test, cross-split domains, PhiUSIIL, D and the complete discovered historical exposure roster. The cohort is locally domain-disjoint, not established as campaign-independent, temporally new or globally unexposed.

The fixed eligible yield is 1,392 positive and 1,262 negative domains. Every eligible domain contributes one representative: the URL with the smallest fixed-prefix SHA-256 ranking, with deterministic ties. A separate domain hash fixes evaluation order. This domain-balanced retrospective estimand differs from the original row-weighted external study. Exact publisher url_norm is the model input; canonical identities serve only eligibility and overlap checks. No predictor output determines inclusion, no new domain is added after scoring, and these labels do not select a model, threshold or gate.

3.16.2 Routing Comparisons and Paired Inference

R scores the accepted structural model and character transformer once per record using their frozen singleton scoring convention and thresholds. The transformer threshold is 0.033976949751377106. Saved component decisions define structural-only, always-transformer, the original zero-width cascade, a symmetric half-width 0.20 gate, and a directional 0.20 band below the structural threshold. The symmetric rule is the selected primary revised gate; the directional rule is a secondary comparison. Both were selected from the bounded development replay in Section 4.8.1 before R predictions. Fixed identity-hash controls select the same number of requests as each revised rule without using correctness. These controls are finite-batch comparisons, not online routers.

The three primary R statements concern original-cascade recall noninferiority to the transformer, symmetric-cascade recall noninferiority to the transformer, and

symmetric-cascade recall improvement over structural-only accompanied by observed FPR at most 1% and calls at most 30%. Noninferiority retains the two-percentage-point margin. Each contrast uses the conservative two-sided 95% discordance envelope in Section 3.8; Newcombe's (1998) method 10 is a prespecified sensitivity. These are individual, conditional intervals rather than simultaneous coverage of all comparisons. Counts, logical calls, negative and positive repairs/harms, and ranking measures are retained for every system.

The sample size is the fixed eligible yield, not a target enlarged until a decision passes. Planning at 1,392 independent positive pairs gives approximately 98.25% power under the exposed internal symmetric-rule discordance proportions. Sensitivity falls to 53.76% at equal recall and 5% total discordance, illustrating that the 90% planning target is scenario-dependent. The actual decision depends on observed paired outcomes and assumptions, not planning power. Positive source-event-date groups are also resampled in 2,000 fixed PCG64 replicates as a descriptive dependence sensitivity. Date groups are not verified campaigns, and a degenerate bootstrap does not supersede the exact envelope.

3.16.3 Actual Detector Work and Audited Controls

C fixes 10 positive and 990 negative R strings in cohort identity order before seeing their scores. The 1% mixture is a workload reference, not estimated traffic prevalence. Six scoring paths—structural, transformer, original cascade, symmetric, directional and symmetric matched-count control—each execute five repetitions of the same 1,000 measured requests at concurrency one. Each arm first executes 100 excluded warmups. Five cyclic rotations of the six-path order are not fully position-balanced. Models load outside the timed region; request timing includes input processing, gate

work and any actual expert scoring, but excludes evidence writes and admission checks. Counts distinguish measured requests from warmups. This is not an HTTP service benchmark.

C retains p50, p95 and p99 for each arm, the summed request-scoring times, scores, decisions and actual stage starts/completions. Within each repetition, summed scoring time is divided by the structural-path total; the five paired ratios are summarized by their median and range. No run-level confidence interval or production-capacity estimate is claimed. Output parity is checked against R's saved component scores. An always-transformer path exercises expert computation separately from natural routing; it is a cost reference, not representative selective traffic.

P fits the prescribed Logistic-L1 pipeline once for each seed 42–46 on the unchanged training identities, shuffling only the ordered training label vector. At the estimator.fit boundary, the retained receipt binds that exact consumed vector, row order, class counts, permutation and transformed input. Reloaded fitted state and saved validation predictions are checked independently. The unchanged 32,695-row validation set has 12,486 positives and 20,209 negatives; its true labels select each descriptive operating point with the existing Clopper–Pearson constraint. Thus, reported validation recall is not independent test efficacy. The five runs are diagnostic controls, not a formal permutation significance test (Ojala & Garriga, 2010). They neither select a favorable seed nor replace the historical fits.

Chapter 4—Results

The results are presented in the order of the research argument. Sections 4.1–4.7 establish the evaluation populations, answer the three original questions and adjudicate every primary requirement. Secondary analyses distinguish operating-point behavior from ranking, calibration and sensitivity. Sections 4.8–4.9 then report the representation and service modifications motivated by the initial findings. The original study and follow-up comparisons retain separate populations, estimands and decision rules.

4.1 Completed Evaluation and Populations

The primary evaluation is complete: both source evaluations, 125 operational cells in 25 five-repeat groups and all 22 hypothesis checks are reported. The accepted two-session matrix retains 72 eligible complete cells from attempt four and 53 continuation cells in their original order. Eligibility, rather than measured performance, determined retention. The secondary program reports its computed analyses together with the source-identifiability assessment and the evidentiary status of the permutation runs. The corrected analysis retains eight passing component checks: five for H1, one for H2 and two for H3. The H3 recall-margin check changes from pass to not established because the earlier zero-width interval did not represent uncertainty beyond the observed pairs. Each joint hypothesis remains unsupported because its rule requires every component to pass. The following sections explain those decisions through detection counts, paired contrasts, monitoring outcomes and service measurements.

Saved-evidence checks reconstruct the original confusion counts, paired outcomes, complete operational groups and pooled latency calculations. All accepted source and service-worker exits are accounted for. The detailed verification counts and

identities are retained in Appendix C rather than used as a measure of scientific contribution.

The accepted continuation completed on October 1, 2026. Result synthesis reads retained observations without fitting or generating new predictions. The post-review analysis corrects paired uncertainty as described in Section 3.8 and leaves the original observations intact.

PhiUSIIL preparation retained 233,536 of 235,795 rows across 197,105 registrable domains. The 2,259 exclusions comprise 1,380 invalid or unsupported uniform resource locators (URLs), 877 same-label canonical duplicates and two rows from one conflicting canonical group. The train, validation and group-test partitions contain 166,248, 32,695 and 34,593 rows, respectively, with 137,973, 29,566 and 29,566 domains. The group-test has 14,326 positives and 20,267 negatives. Its positive stratum contains 9,757 domains. These are publisher reference outcomes, not newly adjudicated labels.

The amended external preparation retained 8,701 of 8,941 published test records, with 240 test exclusions and 4,150 distinct retained domains. The primary outcome strata comprise 69 National Cyber Security Center (NCSC) gold-positive records in 69 domains and 2,497 certified-registry negative records in 234 domains. Secondary strata comprise 417 NCSC silver positives in 412 domains, 4,555 Chongluadao/OpenPhish bronze positives in 2,294 domains, and 1,163 label-free Tranco controls in 1,163 domains. Stratum domain counts need not add to the overall distinct-domain count. The entire retained external stream participates in routing before these outcome strata are applied.

The source contingency is shown below; it must not be interpreted as deployment prevalence.

Table 4.1. Retained external source, tier and outcome composition.

Source / tier	Role	Rows	Domains	Reference outcome
chongluadao openphish / bronze	secondary	4555	2294	phishing (1)
ncsc / gold	gold	69	69	phishing (1)
ncsc / silver	secondary	417	412	phishing (1)
tranco / control	tranco	1163	1163	unlabeled control
trusted registry / certified	certified	2497	234	legitimate (0)

The complete preparation package covers 53,116 publisher rows; its overall 38 invalid-URL and 1,423 PhiUSIIL-overlap exclusions are not test-only counts. Test exclusions, split membership, canonical-overlap and registrable-domain-overlap checks remain in the authenticated preparation records. The original raw-field hold and the exact publisher-url_norm amendment are disclosed in Section 4.6. No local repair or label reinterpretation was used to improve these results.

4.2 RQ1: Representation Value and Generalization

Table 4.2 reports fixed operating points, including the Gaussian mixture model (GMM) routing policy. True positives (TP), false positives (FP), true negatives (TN) and false negatives (FN) denote counts against the source reference labels. Internal recall uses P=14,326 and false-positive rate (FPR) uses N=20,267. External recall uses the 69 gold positives; external FPR uses the 2,497 certified negatives. TN and FN are supplied

explicitly. Upper95 is the exact one-sided 95% Clopper–Pearson FPR bound. The prespecified test gates use observed FPR, not that upper bound. The bounds describe the binomial calculation and do not account for residual dependence among URLs sharing a domain.

Table 4.2. Primary detection counts and rates at unchanged validation-selected thresholds.

Table 4.2a. Confusion counts; P and N are the declared stratum denominators.

Population / model	**TP/P**	**FN**	**FP/N**	**TN**
Internal: Length-only	4958/14326	9368	119/20267	20148
Internal: Logistic-L1	14137/14326	189	157/20267	20110
Internal: Transformer	14229/14326	97	205/20267	20062
Internal: Fixed cascade	14137/14326	189	157/20267	20110
External: Length-only	16/69	53	236/2497	2261
External: Logistic-L1	69/69	0	2261/2497	236
External: Transformer	69/69	0	2479/2497	18
External: Fixed cascade	69/69	0	2261/2497	236
External: GMM policy	69/69	0	2479/2497	18

Table 4.2b. Observed rates and historical row-wise binomial upper bounds. The latter do not establish population coverage for dependent URLs.

Population / model	Recall	FPR	FPR upper95
Internal: Length-only	34.6084%	0.5872%	0.6834%
Internal: Logistic-L1	98.6807%	0.7747%	0.8838%
Internal: Transformer	99.3229%	1.0115%	1.1349%
Internal: Fixed cascade	98.6807%	0.7747%	0.8838%
External: Length-only	23.1884%	9.4513%	10.4699%
External: Logistic-L1	100.0000%	90.5487%	91.4961%
External: Transformer	100.0000%	99.2791%	99.5336%
External: Fixed cascade	100.0000%	90.5487%	91.4961%
External: GMM policy	100.0000%	99.2791%	99.5336%

Structural features increased recall over length alone by 64.0723 percentage points internally and 76.8116 points on gold positives. The internal domain-clustered interval and corrected external independent-pair envelope both exclude zero, conditional on their respective sampling assumptions. The internal structural operating point retained FPR below 1%. On certified external negatives, however, FPR was 90.5487%, compared with 9.4513% for length alone. Thus, the recall gain did not transfer as an acceptable low-FPR operating point. Differences in source composition and input representation prevent attribution of this degradation to a single feature or collection source; ranking and calibration are examined separately in Section 4.5.

Table 4.3. All six primary paired recall contrasts after the uncertainty correction. Differences and intervals are percentage points; methods and dependence assumptions are specified in Section 3.8.

Positive stratum / contrast	Candidate / reference TP	Rows / domains	Difference (pp)	95% interval (pp)
gold.cascade minus logistic l1	69 / 69	69 / 69	0.0000	[-6.1533, 6.1533]
gold.cascade minus transformer	69 / 69	69 / 69	0.0000	[-6.1533, 6.1533]
gold.logistic l1 minus length only	69 / 16	69 / 69	76.8116	[57.2709, 87.1857]
gold.policy minus cascade	69 / 69	69 / 69	0.0000	Not established
internal.cascade minus logistic l1	14137 / 14137	14326 / 9757	0.0000	Not established
internal.logistic l1 minus length only	14137 / 4958	14326 / 9757	64.0723	[59.9897, 67.6302]

The fixed band selects zero of 34,593 internal and zero of 8,701 external rows. Consequently, cascade and Logistic-L1 decisions coincide in both populations; both observed incremental recall differences are zero. The corrected gold interval is [-6.1533, 6.1533] percentage points. The internal degenerate bootstrap establishes no population lower bound; neither contrast establishes the required strict improvement. This is a degenerate selective policy on the evaluated populations, not evidence that character representations are unnecessary generally. Transformer-only recall is 99.3229% internally, versus 98.6807% for Logistic-L1, but its internal FPR is 1.0115% and its

external certified FPR is 99.2791%. That descriptive comparator is not a substitute H1 contrast.

H1 has five passing and five failing components. Its two positive structural-recall contrasts cannot compensate for three failed external FPR checks and two failed incremental-cascade contrasts. H1 is not supported. The answer to RQ1 is therefore conditional: the structural representation adds recall, especially internally, but selective character escalation adds no measured recall in this frozen cascade and the required external low-FPR generalization does not hold.

4.3 RQ2: Monitoring and Future-Only Routing

All six GMM candidates converged under the frozen training procedure. Minimum training Bayesian information criterion (BIC) selected six components, without using external outcomes. Calibration used 16,325 validation rows and audit used 16,370; each stream has 14,783 disjoint domains and 252 complete windows. The GMM boundary remains -67.45792380813624. The original audit alerts on 28/252 windows (11.1111%), exceeding the 5% criterion. That failure was recorded before external evaluation and remains decisive for H2.

Table 4.4. Training-only GMM model-selection record.

Components	Training BIC	Iterations	Converged
1	12267201.154216	2	True
2	-11632435.873684	10	True
3	-20139406.839497	11	True
4	-20182632.805988	11	True
5	-18178918.655785	12	True
6	-22536770.082576	13	True

The external stream yields 132 complete 256-row windows at stride 64. GMM alerts on 115 (87.1212%), passing the 80% detection gate. This measures departure of the external P(X) representation from the fitted reference, not the sensitivity of a detector for independently labeled harmful-drift events. An alert ending at t affects only t+1 through t+256; activations are unioned and truncated at the end of the stream. Incomplete terminal windows do not enter the alert denominator, although their requests can remain subject to earlier alerts.

The saved external policy routes 8,253/8,701 requests (94.8512%) to the transformer, compared with zero fixed-band selections. Gold recall remains 69/69 for both the policy and fixed cascade: the observed paired recall difference, equivalently the reduction in false-negative rate, is zero. The historical zero-width bootstrap interval is not retained as a population bound for this stateful policy. Certified false positives increase from 2,261 to 2,479, a difference of 218 records or 8.7305 percentage points. These are

realized paired policy outcomes, not a randomized causal effect. High alert frequency supplies no evidence that escalation remedies the relevant detection problem.

Table 4.5 compares the GMM with maximum mean discrepancy (MMD) and population stability index (PSI) diagnostics.

Table 4.5. Frozen GMM and descriptive comparator monitors. All rates use complete overlapping windows, without independent-binomial intervals.

Monitor	Fixed boundary	Original audit alerts	External alerts
GMM	-67.45792380813624	28/252 (11.1111%)	115/132 (87.1212%)
MMD	0.01662655786449553	8/252 (3.1746%)	132/132 (100.0000%)
PSI	0.12001239829542137	11/252 (4.3651%)	132/132 (100.0000%)

MMD and PSI each flagged every external window and had lower original audit alert fractions than the GMM. Each used its own training-fixed reference and calibration boundary, so these descriptive comparisons do not change the primary monitor or the decision for H2. The supplement retains all 396 external monitor-window records and 3,432 PSI feature scores. Only the primary external-window alert requirement was met; the audit, certified FPR and incremental gold-recall requirements were not. The answer to RQ2 is therefore specific: the GMM detected the prescribed external departure, but its reference false-alert rate and the low-FPR utility of its routing policy did not satisfy the joint constraints.

4.4 RQ3: Real-HTTP Cost and Inline Viability

The service evaluation measures the Hypertext Transfer Protocol (HTTP) request path rather than model scoring in isolation.

The measured host is an Apple M4 Max MacBook Pro with 16 central processing unit (CPU) cores and 128 gigabytes (GB) of memory, running macOS 15.6.1; the frozen runtime uses singleton transformer inference and the recorded numerical-library limits. Hardware and environment bindings are retained with each accepted segment. The HTTP harness includes serialization, queueing, model execution, body reading and response validation. Warmup, loading and control work are excluded from measured client-phase throughput and latency; phase drain is reported separately. Closed-loop latency excludes unsent local backlog, so these results do not establish open-loop production capacity.

The primary invocation cell recorded zero physical transformer attempts among 10,000 measured requests, satisfying the <=30% requirement and agreeing with the zero fixed-band selection count. This establishes avoided transformer work on that manifest, not preservation of acceptable external detection performance. For cells 21–25, the 50,000 terminal latencies pooled to p95=364.3004101 ms, exceeding 200 ms; request errors were 0/50,000, satisfying the strict <0.1% requirement. These designated primary cells were retained unchanged from the historical session, rather than replaced by later measurements.

Table 4.6. All 25 operational groups. Each row pools five complete repeats. Fixed and transformer rows contain 50,000 requests; live shift contains 43,505. Physical attempts are transformer forward attempts divided by all measured client requests, not a latency reduction.

Workload / concurrency	p50 ms	p95 ms	p99 ms	Errors / requests	Physical attempts
Fixed 1% / c1	1.274	1.489	1.694	0/50000	0.0000%
Fixed 1% / c8	5.754	10.026	13.461	0/50000	0.0000%
Fixed 1% / c16	18.482	45.579	65.890	0/50000	0.0000%
Fixed 1% / c32	43.103	130.176	196.796	0/50000	0.0000%
Fixed 1% / c64	103.386	364.300	581.587	0/50000	0.0000%
Fixed 1% / c128	237.491	897.336	1452.179	120/50000	0.0000%
Fixed 0.1% / c1	1.264	1.480	1.646	0/50000	0.0000%
Fixed 0.1% / c8	5.767	10.037	13.618	0/50000	0.0000%
Fixed 0.1% / c16	18.703	45.351	65.702	0/50000	0.0000%
Fixed 0.1% / c32	45.833	138.859	212.348	0/50000	0.0000%
Fixed 0.1% / c64	102.178	341.839	528.288	0/50000	0.0000%
Fixed 0.1% / c128	222.653	795.941	1217.360	36/50000	0.0000%
Fixed 5% / c1	1.272	1.487	1.654	0/50000	0.0000%
Fixed 5% / c8	5.926	10.310	13.935	0/50000	0.0000%
Fixed 5% / c16	18.136	44.861	68.399	0/50000	0.0000%

Workload / concurrency	p50 ms	p95 ms	p99 ms	Errors / requests	Physical attempts
Fixed 5% / c32	45.369	137.899	207.373	0/50000	0.0000%
Fixed 5% / c64	102.126	338.414	518.365	0/50000	0.0000%
Fixed 5% / c128	227.824	831.311	1291.329	59/50000	0.0000%
Transformer 1% / c1	7.142	8.084	8.581	0/50000	100.0000%
Transformer 1% / c8	49.032	52.725	54.584	0/50000	100.0000%
Transformer 1% / c16	98.300	104.864	107.313	0/50000	100.0000%
Transformer 1% / c32	200.785	243.880	266.039	0/50000	100.0000%
Transformer 1% / c64	412.498	476.335	1017.011	74/50000	99.8800%
Transformer 1% / c128	851.982	970.861	2241.199	1612/50000	96.9640%
Live shift external / c1	10.267	11.582	12.422	0/43505	94.8512%

Table 4.7. Five-run throughput and post-phase drain ranges. Full individual run values and counters are in operational-runs.csv; ranges are descriptions, not uncertainty intervals or best-run selection.

Workload / concurrency	Attempts/s range	Successes/s range	Drain ms range
Fixed 1% / c1	758.45–771.82	758.45–771.82	27.533–31.584
Fixed 1% / c8	1243.93–1272.10	1243.93–1272.10	27.903–29.019
Fixed 1% / c16	725.04–747.06	725.04–747.06	28.040–28.777
Fixed 1% / c32	576.54–595.27	576.54–595.27	27.993–30.495
Fixed 1% / c64	398.91–502.69	398.91–502.69	27.713–30.146
Fixed 1% / c128	308.03–453.34	305.38–453.29	28.497–29.621
Fixed 0.1% / c1	763.33–774.34	763.33–774.34	27.876–29.034
Fixed 0.1% / c8	1244.78–1260.67	1244.78–1260.67	28.355–29.591
Fixed 0.1% / c16	699.52–750.18	699.52–750.18	28.396–28.985
Fixed 0.1% / c32	528.15–582.76	528.15–582.76	28.546–31.158
Fixed 0.1% / c64	435.98–530.84	435.98–530.84	28.656–30.322
Fixed 0.1% / c128	400.75–455.73	400.27–455.60	28.421–30.978
Fixed 5% / c1	761.24–771.45	761.24–771.45	26.976–28.578
Fixed 5% / c8	1172.44–1236.35	1172.44–1236.35	27.884–29.053
Fixed 5% / c16	727.63–752.44	727.63–752.44	28.150–29.277
Fixed 5% / c32	514.84–590.11	514.84–590.11	27.096–29.080

Workload / concurrency	Attempts/s range	Successes/s range	Drain ms range
Fixed 5% / c64	455.18–507.15	455.18–507.15	28.340–30.497
Fixed 5% / c128	394.32–446.43	392.67–446.30	27.862–31.244
Transformer 1% / c1	140.08–141.29	140.08–141.29	27.084–28.343
Transformer 1% / c8	162.21–163.13	162.21–163.13	27.221–28.556
Transformer 1% / c16	162.35–162.90	162.35–162.90	27.360–30.154
Transformer 1% / c32	141.74–163.30	141.74–163.30	28.055–30.852
Transformer 1% / c64	141.00–155.24	140.76–155.04	30.785–32.682
Transformer 1% / c128	136.12–150.23	133.22–145.22	36.661–38.075
Live shift external / c1	97.70–100.76	97.70–100.76	23.550–25.804

The throughput pattern is nonmonotonic. In the 1% fixed workload, concurrency 8 yields approximately 1,244–1,272 attempted requests/s, whereas concurrency 64 yields approximately 399–503. At concurrency 128, fixed-workload request errors occur at all three prevalences (120 at 1%, 36 at 0.1%, and 59 at 5%, each out of 50,000). Transformer-only has 74 errors at concurrency 64 and 1,612 at concurrency 128, with pooled p95 approximately 476.34 and 970.86 ms. Physical attempt fractions below one in those transformer-only groups reflect request/admission/failure behavior, not selective computation. All request failures remain in their original denominators. Admitted/completed/failed work and successful transformer scores are separately retained.

The five serialized live-shift runs reproduce the offline routing trace exactly, with 41,265 physical attempts among 43,505 measured requests (94.8512%), zero request

errors and pooled p95=11.5816 ms. Their concurrency is one, not 64; their workload, denominator and original order differ from H3's fixed reference. This descriptive result establishes execution of the policy, not compliance with the primary latency gate or production safety.

Externally, both cascade and transformer fail certified FPR and Tranco safeguards. Each alerts on all 1,163 Tranco controls. These are label-free control alert rates, not labeled false-positive rates. Both models detect the same 69 gold positives, but the corrected 95% paired envelope is [-6.1533, 6.1533] percentage points under independent-pair sampling. Its lower bound is below -2 points, so the noninferiority requirement is not established. This does not demonstrate inferiority. H3 has two passing and six unmet gates and is not supported. RQ3 is answered by the joint constraints: low transformer use and a clean primary error count do not establish inline viability when external specificity and the designated latency budget fail.

4.5 Secondary Analyses and Interpretation

The secondary analyses examine discrimination, calibration and prevalence sensitivity while retaining every declared model, including unfavorable comparators. They contain 21 internal detector columns and 22 external columns. Together, the internal population and six external population definitions yield 153 population–detector records.

Full mixed-class metrics apply to 43 records: 21 internal and 22 gold-plus-certified. Those records supply 430 calibration bins, 43 low-FPR score curves and 129 prevalence projections. Gold and certified observations also retain their separate

confusion-count strata. Silver and bronze results report positive recall by source and tier; Tranco results report only label-free alerts.

Single-class strata retain the measures their labels support; undefined AP, ROC AUC or calibration summaries are left undefined rather than assigned artificial values. The descriptive summaries also include Matthews correlation coefficient (MCC) and expected calibration error (ECE) where defined. Source identifiability is assessed separately. A source-only classifier would require per-record collection provenance independent of class, which the PhiUSIIL metadata do not supply. Defining source from the label would make the comparison circular. The formatting comparator and external evaluation therefore test representation dependence and transfer, without purporting to isolate a collection-source effect.

Table 4.8. Selected full-metric summaries. External AP and calibration describe the observed 69-positive/2,497-negative mixture, not production prevalence. All columns, precision, F2, MCC and balanced accuracy remain in secondary-metrics.csv.

Population / detector	**AP**	**ROC AUC**	**Brier**	**10-bin ECE**
Internal: Length-only	0.736846	0.712485	0.195827	0.116636
Internal: Logistic-L1	0.995978	0.994856	0.008240	0.009815
Internal: Transformer	0.998310	0.998042	0.003707	0.000514
Internal: Fixed cascade	0.995978	0.994856	0.008240	0.009815
Internal: Formatting	0.768122	0.802108	0.131705	0.052188
Internal: Random Forest	0.996749	0.996394	0.004519	0.002926

Population / detector	AP	ROC AUC	Brier	10-bin ECE
Gold + certified: Length-only	0.218936	0.383893	0.323470	0.517273
Gold + certified: Logistic-L1	0.314882	0.851239	0.876785	0.879966
Gold + certified: Transformer	0.099927	0.592273	0.965747	0.965905
Gold + certified: Fixed cascade	0.314882	0.851239	0.876785	0.879966
Gold + certified: Formatting	0.026890	0.494994	0.103375	0.268231
Gold + certified: Random Forest	0.036487	0.634068	0.867826	0.879992
Gold + certified: GMM policy	0.175659	0.498204	0.965747	0.965902

Internal Logistic-L1 AP is 0.995978, but external AP is 0.314882 and ECE is 0.879966. External transformer and policy ECE are approximately 0.9659. Even an external ranking statistic above chance cannot make the frozen operating threshold acceptable. The descriptive low-FPR score curves enumerate tied score cutoffs without interpolation; they are not newly selected deployment thresholds. For example, the structural external curve attains 20/69 positives with 24/2,497 false positives at a score cutoff near 0.999992. That retrospective curve must not replace the frozen operating result of 2,261 false positives.

Prevalence projections transport the measured class-conditional true-positive rate (TPR) and FPR to 0.1%, 1% and 5% hypothetical phishing prevalence. They are not new HTTP measurements. At 1%, the external fixed-cascade rates imply approximately 9,064 alerts per 10,000, including 8,964 false alerts and zero misses under that assumption. This arithmetic illustrates the cost of poor specificity; it neither validates the transport

assumption nor establishes actual deployment prevalence. All originating rates and projection values are supplied.

Table 4.9. Additional positive tiers and label-free controls at frozen primary thresholds.

Detector	**Silver TP / 417**	**Bronze TP / 4,555**	**Tranco alerts / 1,163**
Length-only	18 (4.3165%)	35 (0.7684%)	0 (0.0000%)
Logistic-L1	417 (100.0000%)	4555 (100.0000%)	1163 (100.0000%)
Transformer	417 (100.0000%)	4554 (99.9780%)	1163 (100.0000%)
Fixed cascade	417 (100.0000%)	4555 (100.0000%)	1163 (100.0000%)
GMM policy	417 (100.0000%)	4554 (99.9780%)	1163 (100.0000%)

The five-indicator formatting comparator attains internal recall 60.4216% with zero observed false positives, but external gold recall is zero and certified FPR is 1.0012%. This shows source-dependent predictive structure in a restricted representation; it is not a causal shortcut diagnosis. The fixed 100-tree Random Forest achieves internal recall 99.2042%, FPR 0.7006%, AP 0.996749 and AUC 0.996394. Externally, recall is 100% but FPR is 91.3496%, AP 0.036487 and AUC 0.634068. Its correction-v2 artifact and validation cutoff 0.2 are retained; it is a secondary benchmark, not a replacement primary model.

Table 4.10. Every accepted permutation comparator; no seed selection.

Permutation seed	Internal AP	Internal AUC	External AP	External AUC
42	0.533318	0.481551	0.025910	0.369989
43	0.672760	0.624953	0.272610	0.827396
44	0.551078	0.546285	0.017823	0.176574
45	0.437844	0.402251	0.128390	0.837681
46	0.772821	0.687614	0.190236	0.738530

The historical permutation table remains a sensitivity record. Its reconstructed label vectors preserve training class counts, but no retained digest links those vectors to the labels actually consumed by the historical estimators. Score means near 0.5 cannot establish that linkage or explain ranking variation. Section 4.10.4 reports all five new, separately documented fits with direct consumed-label evidence. Those fits do not retroactively authenticate the historical ones and neither family establishes the presence or absence of leakage.

Table 4.11. Transformer seed/runtime sensitivity at each accepted secondary operating point. T denotes transformer. All five secondary cascades coincide with Logistic-L1 on these evaluated populations.

Seed	Internal T recall	Internal T FPR	Gold T recall	Certified T FPR	Fixed band selections
42	99.3229%	1.0115%	100.0000%	99.2791%	0 internal; 0 external
43	99.3159%	0.7253%	100.0000%	99.3192%	0 internal; 0 external
44	99.3369%	0.8783%	100.0000%	99.3192%	0 internal; 0 external
45	99.3299%	1.0362%	100.0000%	99.3592%	0 internal; 0 external
46	99.3020%	0.7549%	100.0000%	99.3192%	0 internal; 0 external

Seeds 42–46 remain a single accepted family. Seed 42 used its historical training runtime; seeds 43–46 used the later pinned runtime, so their differences are not an isolated causal seed effect. Each had three logical band selections among 32,695 validation rows and zero selections in the internal and external evaluation streams. The full accepted cutoffs, bands, checkpoint records and development values are retained in complete-secondary-results.json. No best seed is promoted.

Table 4.12. Positive-only paired McNemar tables and four-slot Holm adjustment. “Underflow” means the saved floating-point computation returned 0.0, not an exact mathematical p-value of zero.

Contrast	Both correct	Candidate only	Reference only	Both incorrect	Exact p	Holm p
internal logistic minus length	4952	9185	6	183	underflow	underflow
internal cascade minus logistic	14137	0	0	189	1	1
external gold logistic minus length	16	53	0	0	2.22045e-16	6.66134e-16
external gold cascade minus logistic	69	0	0	0	1	1

These nominal paired tests use aligned positive outcomes and do not model registrable-domain or routing dependence. They supplement, rather than decide, the primary clustered contrasts. A small p-value for structural recall does not cancel an external FPR failure.

Table 4.13. Accepted label-free development probe accounting.

Stream	Rows	Eligible	Changed	Eligible no-op	Ineligible
original	16370	16370	0	16370	0
ascii scheme host uppercase	16370	16370	16370	0	0
percent escape hex uppercase	16370	53	22	31	16317
first literal path alphanumeric percent encode	16370	1563	1563	0	14807

All four probe streams retain 16,370 row positions and 252 complete overlapping monitor windows. American Standard Code for Information Interchange (ASCII) scheme/host uppercasing produces no binary detector transitions, but GMM alerts increase from 28 to 29 and MMD from 8 to 15, while PSI decreases from 11 to 10. Percent-escape uppercasing changes only 22 eligible records and produces no binary or monitor-alert transitions. It is therefore weak evidence for any broad invariance claim. First-literal path encoding changes 1,563 records: GMM-policy decisions change $0 \rightarrow 1$ for 105 and $1 \rightarrow 0$ for 63; length-only has 48 new alerts; transformer has 14; fixed cascade and Logistic-L1 have none. GMM alerts increase to 213, MMD to 14 and PSI decrease to 6. Full paired score deltas, all four decision-table cells, monitor transitions, fixed boundaries and eligibility counts are supplied in probe-decisions-and-scores.csv and probe-monitors-and-scores.csv. These streams have no inherited outcome labels and

establish no accuracy, semantic-equivalence, adversarial-success or production-robustness rate.

4.6 Evaluation Provenance and Information Exposure

Model acceptance, information exposure and session structure affect how these results can be interpreted. The accepted models and thresholds are fixed during evaluation. Appendix C retains the failed predecessors and their corrections, keeping execution repair distinct from evidence of detection performance.

External preparation initially found no eligible gold or Tranco population because the raw publisher values lacked schemes. Adopting the publisher's exact url_norm for parsing and model input restored the specified populations while preserving raw values, labels, overlap rules and model thresholds. This choice followed preparation and label-count inspection but preceded amended predictions. It was therefore a disclosed representation decision, not a wholly pre-access specification. Appendix C gives the preparation counts and amendment record.

The operational matrix uses the complete 72-cell prefix from one selected attempt and the subsequent 53-cell continuation. Selection required the entire prefix and both source outputs to pass an eligibility review. Partial requests were excluded, and earlier attempts supplied no substitute measurements. The two-session consequence is addressed in Section 3.12; Appendix C retains each interruption and its exclusion.

The resulting matrix is a two-session evaluation, a departure from the original single-session design. Descriptive group 71–75 crosses the boundary, while the designated primary cells retain their original observations. Thermal state, caches, background activity and elapsed time can therefore covary with the fixed workload order;

session labels alone do not identify their separate effects. Environmental checks are sampled records rather than continuous experimental control. Source evaluation also involved repeated scoring, and broad searches on September 3 and September 9 exposed test-row content to the analyst. The models and selection thresholds remained frozen, but the evaluation population is neither lifetime-unseen nor evaluated in only one lifetime pass.

4.7 Complete Hypothesis Adjudication

Table 4.14 contains every primary gate. For a contrast row, “Observed operand” is the lower confidence endpoint used in the corrected decision, not its point estimate. Where dependence or a degenerate bootstrap leaves a valid lower bound unestablished, that status is shown rather than population certainty at zero; Table 4.3 gives both. Observed rate counts and historical binomial bounds are in Table 4.2 and the accompanying primary-gates.csv. All decisions use full-precision values; rounded displays never determine passage. The H3 latency and error denominators are the designated five primary runs, not all 125 cells.

The revision's corrected-primary-gates.csv records both original and corrected decisions; the previously released primary-gates.csv preserves the original analysis. Historical row-wise binomial bounds carry the dependence qualification in Section 3.8. No FPR threshold is changed by the correction.

Table 4.14. All 22 primary hypothesis checks with unchanged numerical requirements and corrected paired inference. A failed inferential check means its requirement is not established.

H	Gate	Observed operand	Required	Result
H1	internal.length only.fpr	0.5872%	<= 1.0000%	pass
H1	internal.logistic l1.fpr	0.7747%	<= 1.0000%	pass
H1	internal.cascade.fpr	0.7747%	<= 1.0000%	pass
H1	certified.length only.fpr	9.4513%	<= 1.0000%	fail
H1	certified.logistic l1.fpr	90.5487%	<= 1.0000%	fail
H1	certified.cascade.fpr	90.5487%	<= 1.0000%	fail
H1	internal.logistic l1 minus length only	59.9897 pp	> 0.0000 pp	pass
H1	internal.cascade minus logistic l1	Not established	> 0.0000 pp	fail
H1	gold.logistic l1 minus length only	57.2709 pp	> 0.0000 pp	pass
H1	gold.cascade minus logistic l1	-6.1533 pp	> 0.0000 pp	fail
H2	external window alerts	87.1212%	>= 80.0000%	pass
H2	audit window alerts	11.1111%	<= 5.0000%	fail
H2	certified.policy.fpr	99.2791%	<= 1.0000%	fail
H2	gold.policy minus cascade	Not established	> 0.0000 pp	fail
H3	certified.cascade.fpr	90.5487%	<= 1.0000%	fail
H3	certified.transformer.fpr	99.2791%	<= 1.0000%	fail

H	Gate	Observed operand	Required	Result
H3	tranco.cascade.alert rate	100.0000%	<= 1.0000%	fail
H3	tranco.transformer.alert rate	100.0000%	<= 1.0000%	fail
H3	gold.cascade minus transformer	-6.1533 pp	>= -2.0000 pp	fail
H3	reference transformer invocations	0.0000%	<= 30.0000%	pass
H3	http pooled p95 ms	364.3004 ms	<= 200 ms	fail
H3	http request errors	0.0000%	< 0.1000%	pass

H1: 10/10 components measured, five pass, five fail, not supported. H2: 4/4 measured, one passes, three fail, not supported. H3: 8/8 assessed, two pass, six requirements are unmet, not supported. Across the corrected analysis of the original evaluation, eight of 22 checks pass and fourteen do not. One decision changes: the 69-pair recall comparison no longer establishes the two-point noninferiority margin. These decisions apply to the prespecified conjunctions; they do not imply that every component was unsuccessful or that every related model has been rejected. The direct research-question answers follow from the measured effects and constraints, with the secondary identifiability and audit limitations retained in their interpretation.

4.8 Diagnostic Investigation and Detection Comparison

4.8.1 From Aggregate Behavior to Testable Mechanisms

The initial evaluation established structural recall gains with positive clustered lower bounds, internal FPR compliance for the three primary systems, detection of 115 of 132 external windows, and compliance with the reference cascade's invocation and primary error-rate limits. It also identified two questions for diagnostic follow-up:

whether scheme spelling contributed a removable representation dependency, and whether HTTP client organization contributed to the observed latency. These questions motivated targeted comparisons rather than changes to the original decision rules. An adverse outcome alone was not treated as evidence of a software defect or of a particular causal mechanism.

The external publisher representation makes scheme sensitivity a specific testable issue. The original structural extractor uses both is_https and string-composition features. Median external-window PSI for is_https is 5.8348, although other dimensions also shift. This diagnostic neither identifies the cause of external FPR nor establishes independent window-level significance. It motivates a representation change whose intended invariance can be checked exactly. The separate no-model HTTP diagnostic motivates comparison of shared-pool and worker-owned clients; instrumented development timings are excluded from final performance evidence.

The saved decisions identify a precise obstacle to gate-only correction. Among 2,497 certified negatives, both structural and transformer models misclassify 2,261; the transformer repairs none of the structural false positives and introduces 218 additional errors. Both correctly reject only 18. Consequently, even an oracle choosing between their unchanged binary decisions has observed FPR at least 2,261/2,497, or 90.5487%. This bound applies to these operating points and records, not to altered thresholds or retrained models. It explains why routing more requests cannot, by itself, meet the external 1% requirement.

Ranking and operating-point transfer answer different questions. The structural score has internal ROC-AUC 0.994856 and external gold-plus-certified ROC-AUC

0.851239; the transformer's corresponding values are 0.998042 and 0.592273. Even the best retrospectively inspected threshold with observed certified FPR at most 1% detects only 20/69 gold positives for the structural model and 7/69 for the transformer. Thus, a threshold change alone cannot retain their observed perfect gold recall at the intended FPR on these scores. These frontiers use exposed outcome labels for diagnosis; they are not independently validated deployment thresholds.

The original call-minimizing objective also explains the inactive fixed cascade. Its accepted threshold is 0.2670846328466124 and its half-width is zero. Inclusive membership admits three exact threshold ties in validation, but no ties occur in either evaluation cohort. Replaying all 43,294 saved feature vectors, routing memberships and binary decisions reproduces the accepted outputs. The mechanism is a feasible validation rule that collapses to the structural model on the evaluated inputs, rather than an inference that the transformer was unavailable.

A bounded post-review development replay tests whether a less restrictive gate can select useful work. Before replay, the specification fixed symmetric, below-threshold and above-threshold bands with widths 0.025, 0.05, 0.10 and 0.20. Selection used only the already-exposed internal outcomes, with observed FPR at most 1%, recall no more than two points below transformer-only and calls at most 30%. It maximized detected positives, then preferred fewer false positives, fewer calls and the declared width/name tie-breaks. One symmetric and one directional candidate were retained. This is development selection, not untouched confirmation or a replacement for the original H1–H3 decisions.

The selected symmetric band of width 0.20 routes 1,146/34,593 internal records (3.3128%). Relative to structural-only, true positives rise from 14,137 to 14,171 and false positives fall from 157 to 117. A fixed hash-ranked control using the same number of calls yields 14,142 true positives and 166 false positives. The selected below-threshold band uses 1,039 calls and yields 14,172 true positives but 199 false positives. Applied to the exposed external cohort, both selected gates retain 69/69 gold detections and increase certified false positives from 2,261 to 2,272: eleven harms and no repairs. These are logical routing counts, not new timing measurements. The finite-batch hash control is not an online random-routing policy. The result establishes an internal conditional-repair benefit for the symmetric band and shows that this benefit does not transfer to the tested external negatives.

4.8.2 Population Flow and Operating-Point Comparison

Table 4.15 reports population flow. The 2,808 quarantined rows are a row count. The four reason incidences overlap and therefore must not be summed as if they were mutually exclusive exclusions. Complete source-order predictions exist for all 8,622 eligible rows, with no observed scientific execution failure in fitting or external scoring. A separate arithmetic implementation reproduces every saved metric, interval and D decision without refitting a model or generating new predictions.

Table 4.15. Additional benchmark admission; reason incidences are overlapping.

Item	Count	Interpretation
Publisher rows	11,430	5,715 per reference class
Quarantined rows	2,808	Unique excluded rows
PhiUSIIL domain overlap	2,134	Reason incidence, before filtering
PhishVN domain overlap	1,212	Reason incidence, before filtering
Invalid URL	108	Reason incidence
Same-label canonical duplicate	10	Reason incidence
Eligible publisher-phishing	4,651	2,897 registrable domains
Eligible publisher-legitimate	3,971	3,377 registrable domains
All eligible rows	8,622	6,273 domains; largest cluster 351 rows

Table 4.16 gives the comparison at the two frozen thresholds. The candidate produces 251 fewer false positives but 184 fewer true positives than the comparator. These changes are a measured tradeoff, not an unqualified improvement. The original models and original hypothesis decisions are unchanged.

Table 4.16. Additional benchmark detection results; P=4,651 and N=3,971.

Quantity	Unchanged 25-feature model	Transport-neutral 24-feature model
Threshold	0.2670846328	0.6815549263
TP / FN	4,619 / 32	4,435 / 216
FP / TN	3,940 / 31	3,689 / 282
Recall	99.31%	95.36%
FPR	99.22%	92.90%
Precision	53.97%	54.59%
ROC AUC	0.6821	0.6950
Average precision	0.6651	0.6976
Brier score	0.4306	0.4077

The primary recall difference is −3.9561 percentage points, with a 97.5% domain-cluster interval of [−5.0547, −3.0014] points. The candidate FPR interval is [91.8651%, 93.9100%]. Neither interval has undefined resamples among the 10,000 replicates. The largest cluster contains 351 rows, so treating the 8,622 rows as independent would not reflect the specified dependence structure. The ROC AUC, average-precision and Brier differences in Table 4.16 are descriptive; no additional significance claim is assigned to them.

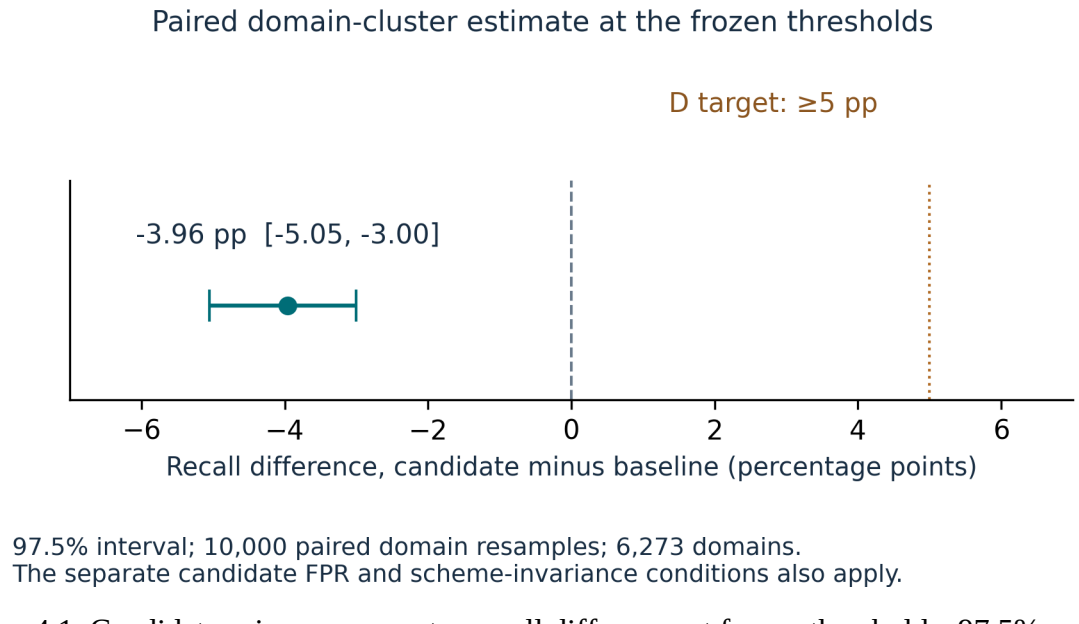
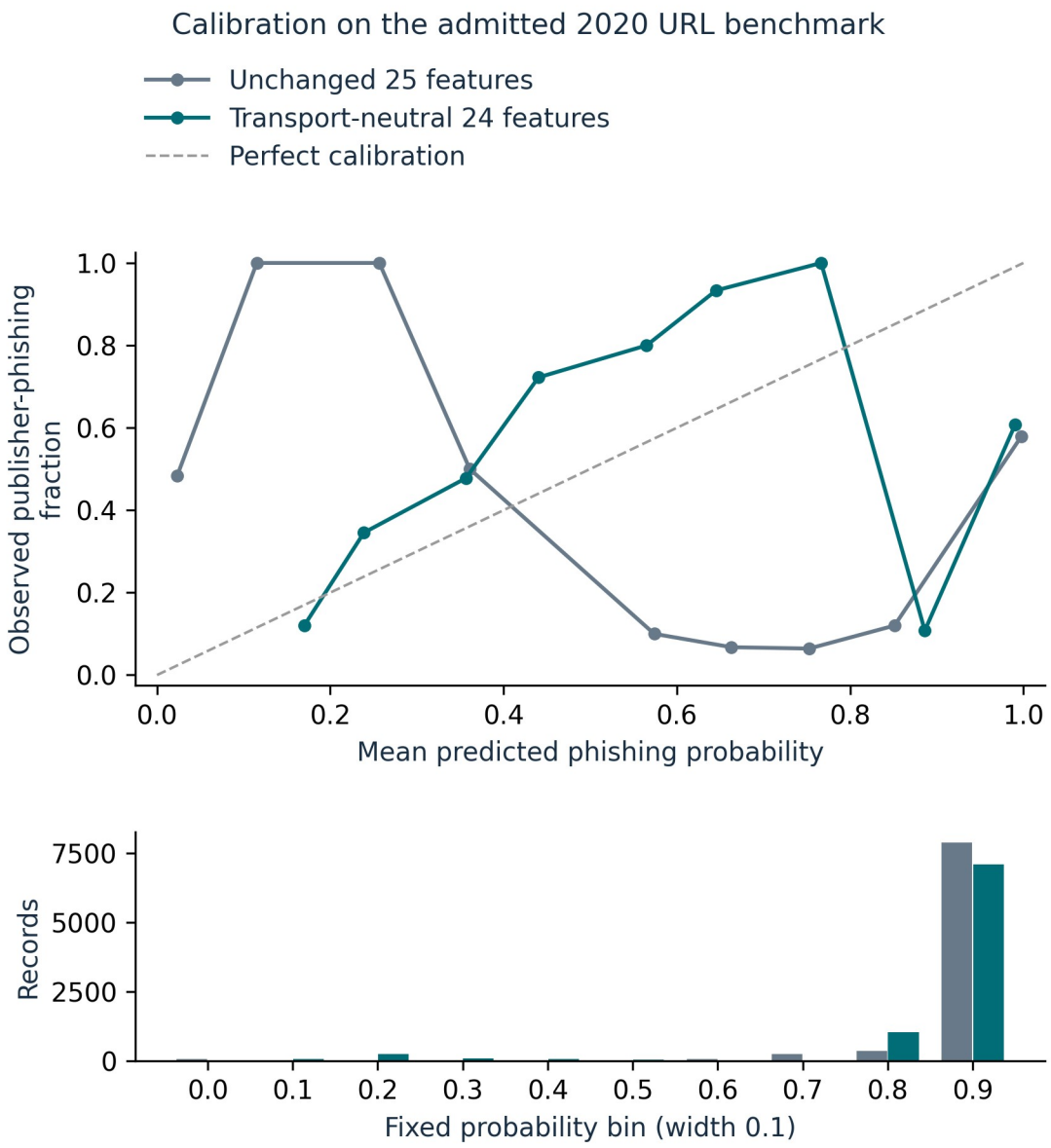


Figure 4.1. Candidate-minus-comparator recall difference at frozen thresholds; 97.5% domain-cluster interval.



8,622 records; observed class mixture, not deployment prevalence.
Empty bins have no reliability point; probability 1 belongs to the final bin.

Figure 4.2. Fixed-bin calibration and bin populations on the admitted benchmark. Small populated bins can have extreme fractions; empty bins have no reliability point. The class mixture is not deployment prevalence.

4.8.3 Representation Invariance and Decision

The intended representation property is achieved on every evaluated pair. All 8,622 candidate feature vectors and scores match their opposite-scheme companions exactly; no candidate decision flips. The comparator has 2,909 exact score matches, 481 decision flips and a maximum absolute score change of 0.99665. A comparator flip is a demonstrated sensitivity to the transformation, not automatically an incorrect classification. The companion test does not supply additional independent labeled examples.

Table 4.17. Prespecified follow-up D requirements, separate from H1.

Requirement	Observed result	Decision
Recall gain at least 5 pp	−3.9561 pp	Not met
97.5% recall-difference lower bound above zero	−5.0547 pp	Not met
Candidate observed FPR at most 1%	92.8985%	Not met
Exact candidate feature/score invariance	8,622 / 8,622; zero decision flips	Met
D conjunction	Three efficacy requirements not met	Not supported

The modification removed the measured scheme-spelling dependency but did not satisfy the low-FPR transfer requirement. This narrows the interpretation of the proposed remedy: exact scheme invariance was achievable, yet insufficient for the fitted structural model to meet D on the admitted benchmark. The comparison does not establish that scheme information is universally irrelevant or identify every remaining source effect. Its

reusable result is an explicit representation contract, accompanied by a measured recall–specificity tradeoff that prevents conformance from being mistaken for detection efficacy.

4.9 Controlled Service Comparison: Completed Evaluation

4.9.1 Workload Scope and Complete Accounting

The final service comparison contains 80 arms: two workloads, two concurrency levels, ten pairs and two client topologies. It contributes 800,000 measured requests and 80,000 excluded warmups. All 80 service processes exited normally. The complete schedule is analyzed separately from the interrupted predecessor described in Appendix C; its results measure the specified synthetic, structural-only closed-loop workload.

The replacement schedule was specified after partial nonprimary observations, before the primary structural concurrency-64 comparison had started. All predecessor observations remain excluded. This timing matters for the study's chronology but does not change which 80 arms supply the estimate.

A separate arithmetic implementation reproduces all 80 arm summaries, 40 paired comparisons, eight pooled groups and five S requirements from saved observations. The final schedule records one shared-client timeout and no warmup errors. The original 125 cells, 25 groups and 1,901 request errors remain a separate experiment. The verification performs no fitting or predictions.

4.9.2 Paired Tail-Latency Effect

Table 4.18 reports every primary pair. Worker p95 is lower than shared-client p95 in each pair. The median worker/shared ratio is 0.203249, equivalent to a 79.68% median

paired reduction. Its two-sided 97.5% run-bootstrap interval is [0.187903, 0.213508], using the fixed 10,000 resamples. No pair or replicate is undefined.

Pooled successful-request p95 is 71.9990 ms for worker clients and 358.3663 ms for shared clients. The worker value meets the 200-ms requirement. These pooled quantiles describe the combined successful requests; their ratio does not replace the prespecified median of paired per-run ratios.

Table 4.18. All ten structural-detector concurrency-64 pairs; success-only p95 in milliseconds.

Pair	Shared p95	Worker p95	Ratio	Exact / requested
1	343.1552	72.6055	0.211582	85 / 10,000
2	371.7900	67.2667	0.180927	83 / 10,000
3	359.2445	70.0094	0.194879	71 / 10,000
4	358.8575	72.8568	0.203024	79 / 10,000
5	397.6750	71.5949	0.180034	89 / 10,000
6	342.2972	74.7204	0.218291	80 / 10,000
7	362.0759	74.3130	0.205242	75 / 10,000
8	352.6524	71.7555	0.203474	84 / 10,000
9	341.7835	73.6319	0.215434	71 / 10,000
10	353.7333	71.2943	0.201548	82 / 10,000

4.9.3 Controls, Throughput, Errors and Response Semantics

Table 4.19 reports all required control/concurrency combinations, not only the primary comparison. Each row pools ten arms and 100,000 measured attempts. Throughput is attempted requests divided by the sum of measured client-phase durations; it excludes separately retained warmup and drain intervals. With zero errors, successful-response throughput equals attempted throughput. The shared structural c64 row instead has 99,999 successes and one error. Successful, failed and all-request percentiles and physical counters are preserved in the aggregate comma-separated values (CSV) files; Appendix B gives every arm.

Table 4.19. Complete pooled service groups; success p50/p95/p99 in milliseconds and attempted throughput in requests per second.

Workload / c / client	p50	p95	p99	Errors	Requests/s
Control / 1 / shared	0.93	1.10	1.24	0	1039.88
Control / 1 / worker	0.93	1.11	1.26	0	1036.52
Control / 64 / shared	103.91	345.95	516.29	0	469.21
Control / 64 / worker	35.86	67.21	77.34	0	1647.64
Structural / 1 / shared	1.17	1.42	1.63	0	820.44
Structural / 1 / worker	1.17	1.43	1.63	0	819.76
Structural / 64 / shared	107.52	358.37	536.11	1	454.14
Structural / 64 / worker	37.36	72.00	85.37	0	1526.71

At concurrency 1, the two topologies have nearly identical pooled p95: 1.10 versus 1.11 ms for the control and 1.42 versus 1.43 ms for the structural workload. At concurrency 64, control p95 falls from 345.95 to 67.21 ms and structural p95 from 358.37 to 72.00 ms. Structural attempted throughput rises from 454.14 to 1,526.71 requests/s; successful shared throughput is 454.14 requests/s when rounded to two decimals. The control pattern supports a client/transport explanation for this intervention's benefit rather than a faster classifier. It does not isolate individual connection-pool, scheduling or queueing mechanisms, and it does not estimate a transformer speedup.

The only request error occurs in structural c64 pair 2's shared arm. Its timeout latency is 2,019.144208 ms; that is the sole observation for all three failure-only quantiles. A two-second deadline can produce a terminal elapsed time slightly above two seconds when cancellation and bookkeeping complete. This is retained as a timeout, not relabeled a success. That arm admits and completes 9,999 measured requests; the timed-out request is not counted as admitted. Across the full schedule there are 799,999 measured admissions/completions, zero server failed-request increments and zero transformer attempts or successful transformer scores. A client timeout and a server failure are different counters.

All 100,000 primary worker requests succeed. One shared-client timeout leaves 99,999 comparable primary request pairs. Every comparable pair agrees exactly on action, probability and stage-2 invocation when request IDs and admission sequence are excluded.

The strict response rule excludes request IDs only, so it also requires equal admission sequences. Under that rule, 799/100,000 primary pairs agree. The other 99,200 comparable pairs differ in admission sequence, not prediction fields; the timed-out pair remains noncomparable.

Across all 40 pairs in the schedule, predictions agree for all 399,999/399,999 comparable request pairs. The same timeout accounts for the one noncomparable pair. Agreement was therefore not observed for all 400,000 requested pairs.

4.9.4 Requirement S and Engineering Interpretation

Table 4.20 separates demonstrated performance from the unchanged response contract. Four of five S requirements pass. S as a conjunction is not supported because full response equivalence includes admission sequence. No threshold or agreement field is removed after seeing this result.

Table 4.20. Prespecified S requirements and verified decisions.

Requirement	Observed result	Decision
Median paired ratio ≤0.8	0.203249	Pass
97.5% ratio upper bound <1	0.213508	Pass
Pooled worker success p95 ≤200 ms	71.9990 ms	Pass
Worker errors <0.1%	0 / 100,000 (0%)	Pass
Exact paired responses except IDs	799 / 100,000	Not met

The modification nevertheless demonstrates a concrete service improvement: the structural HTTP workload meets the tail-latency and reliability requirements with worker-owned clients, while preserving every comparable prediction. This is a measured

transport improvement, not a deployment qualification for the detector and not retrospective support for original H3. Applications that require equivalent admission ordering would need an explicitly designed ordering contract and a new evaluation. Applications that need request-correlated prediction equivalence may consider these results informative, but cannot retroactively substitute that narrower contract for S.

4.10 Post-review Findings: Representation, Routing and Cost

All four specified tasks completed once. The retained evidence independently reconstructs the F transitions, every R decision and routing count, all 30 C arms, and the consumed-label linkage of all five P fits. No fitting or new prediction occurs in the verification. The following results extend the explanation of the original study; they do not revise its populations or numerical requirements.

4.10.1 Frozen-model Scheme Intervention

Table 4.21 shows a large and asymmetric dependence on the prefix supplied to the unchanged structural model. HTTP yields all 69 gold detections but also alerts on every certified negative. HTTPS reduces certified false positives from 2,261 to 50 while reducing gold detections from 69 to 22. In paired terms, that is 2,211 fewer false positives and 47 fewer detected positives. The resulting 2.0024% FPR still exceeds 1%, and recall is only 31.8841%. The result rejects a scheme-prefix replacement as a sufficient remedy for this frozen detector.

Table 4.21. Frozen structural-model scheme intervention. P=69 gold positives; N=2,497 certified negatives. Every character after the scheme colon is preserved.

Input form	TP	FN	FP	TN
Retained spelling	69	0	2,261	236
Forced HTTP	69	0	2,497	0
Forced HTTPS	22	47	50	2,447

The remaining strata show the same tradeoff without changing their evidentiary roles. HTTPS reduces detections among the 4,972 secondary positives from 4,972 to 687 and reduces Tranco alerts from 1,163 to 63. Tranco remains a label-free reference, not a verified negative population. F is distinct from D: F changes input spelling under fixed weights and threshold, whereas D refits and recalibrates a revised representation on the original development partitions. The combined evidence demonstrates scheme sensitivity and a successful invariance property, not a unique causal explanation for every external error.

4.10.2 Expanded Domain-disjoint Routing Evaluation

Every system detects all 1,392 positive representatives. Specificity separates them. Structural-only and the original cascade produce 1,028 false positives among 1,262 negatives (81.4580%). Always-transformer produces 1,221 (96.7512%). The symmetric and directional gates each produce 1,046 (82.8843%): both add 18 new false positives and repair none. Their positive outcomes also contain no repair or harm. Symmetric and directional matched-count controls add only two and one false positives, respectively, although those controls provide no deployable routing rule.

Table 4.22. R cohort decisions and logical expert selections. P=1,392; N=1,262. Every row uses the same one-URL-per-domain cohort; calls are out of 2,654 records.

System	TP	FN	FP	TN	Calls
Structural	1,392	0	1,028	234	0
Original cascade	1,392	0	1,028	234	0
Transformer	1,392	0	1,221	41	2,654
Symmetric 0.20	1,392	0	1,046	216	21
Directional 0.20	1,392	0	1,046	216	18
Symmetric control	1,392	0	1,030	232	21
Directional control	1,392	0	1,029	233	18

For original and symmetric cascades versus transformer-only, each positive paired table has 1,392 concordant detections and zero discordances. The conservative two-sided 95% envelope is [-0.3143, 0.3143] percentage points; its lower endpoint exceeds the unchanged -2-point margin. Both prespecified recall-only noninferiority statements are supported under independent labeled-pair assumptions. Newcombe method 10 gives the approximate sensitivity interval [-0.2752, 0.2752] points. Positive records occupy 56 source-event-date groups, the largest containing 362 records. Their all-zero-difference bootstrap returns [0,0], a descriptive outcome that does not certify campaign independence or replace the nondegenerate paired interval.

The symmetric benefit-and-budget statement is not supported. Calls are 21/2,654 (0.7913%), below 30%, but recall does not improve and FPR is far above 1%. Directional calls are 18/2,654 (0.6782%). Across the full cohort, the transformer repairs one

structural false positive and introduces 194; the selected score bands do not capture that isolated repair. Structural ROC AUC/AP are 0.7549/0.7485, versus 0.7062/0.6909 for the transformer. These descriptive rankings and the near-balanced cohort's AP are not population-prevalence estimates. The new recall precision is a positive result with a narrow meaning: specificity, useful escalation and original joint H3 support do not follow from it.

4.10.3 Measured Detector Computation

C completes all 30 arms and 30,000 measured requests, plus 3,000 excluded warmups. On its fixed workload, the symmetric rule makes 17 actual transformer calls per 1,000 measured requests, the directional rule 15 and the original cascade zero. The symmetric matched-count control also makes 17. Across measured and warmup phases together, 33,000 requests complete without a recorded request failure; actual transformer starts and completions both equal 5,790. The measured-only contrasts use post-warmup counter differences, not that whole-run total.

Table 4.23. Actual detector computation on C's fixed 1,000-string workload. Quantiles are medians of five run quantiles in milliseconds. Ratio is the median paired summed-scoring-time ratio to structural-only, not an HTTP latency ratio.

Scoring path	Calls / 1,000	p50 ms	p95 ms	p99 ms	Time ratio
Structural	0	0.187	0.198	0.226	1.000
Original cascade	0	0.189	0.204	0.245	1.009
Symmetric 0.20	17	0.189	0.223	6.769	1.646
Directional 0.20	15	0.190	0.217	6.802	1.589
Symmetric control	17	0.189	0.233	7.015	1.710
Transformer	1000	5.559	6.657	16.484	30.781

The symmetric ratio is 1.646, with observed run-pair range 1.585–1.743. Directional routing costs 1.589 times structural-only, the matched-count control 1.710, and always-transformer 30.781. The original zero-call cascade remains close to structural-only at 1.009. These are descriptive single-host comparisons with five repeats, not confidence intervals. The symmetric median run p95 is 0.223 ms but p99 is 6.769 ms: its 1.7% expert-use fraction lies beyond the 95th percentile yet enters the 99th-percentile tail. Reporting p95 alone would obscure that cost. Read alongside R's absence of beneficial selected repairs, C gives no measured reason to adopt the wider gate for the tested source frame.

4.10.4 Audited Shuffled-label Controls

All five new P fits preserve class counts and prove that the specified shuffled vector entered estimator.fit in the correct row order. Each fit converges below the fixed 5,000-iteration limit; saved-state and singleton-score checks pass. Table 4.24 reports every seed on the unchanged validation labels. Because those labels also select each threshold, the table is a diagnostic of the stated procedure rather than a held-out performance result.

Table 4.24. All five audited shuffled-label fits. Validation P=12,486; N=20,209. AUC and AP use unchanged validation labels; TP/FP use validation-selected descriptive operating points.

Seed	ROC AUC	AP	TP	FP	Iterations
42	0.4718	0.4871	1,623	178	2001
43	0.6419	0.6718	4,460	175	698
44	0.5487	0.5206	1,250	176	1010
45	0.3940	0.4027	730	178	700
46	0.6965	0.7683	6,882	177	341

The AUCs range from 0.3940 to 0.6965, and AP from 0.4027 to 0.7683; these outcomes are not described as uniformly chance-like. Coefficient-vector L2 norms range from 0.0299 to 0.0851. For every seed, the central half of validation probabilities lies within approximately 0.4977–0.5024, although tail scores extend further. AUC depends on score ordering rather than distance from 0.5: a tightly compressed score distribution can still have a nontrivial ranking. This observation explains why a near-0.5 center

cannot validate a control, but does not establish the unique cause of the seed-specific rankings. True validation-label threshold selection is another reason not to interpret the operating points as independent null-test outcomes.

The audited controls close the consumed-label linkage question for these five fits. They do not retroactively authenticate historical comparators, prove leakage absence or support a formal significance claim from five permutations. No seed was discarded or rerun for a more convenient result. This narrower conclusion is useful: the new evidence distinguishes what entered fitting from what the resulting ranks mean, leaving neither class-count preservation nor an attractive aggregate metric to stand in for direct input verification.

Chapter 5—Discussion and Conclusions

5.1 Direct Answers to the Research Questions

The original study answers all three questions through their declared metrics. After paired-inference correction, eight of 22 checks pass: H1 five, H2 one and H3 two. Joint hypotheses require every constituent condition. The subsequent studies diagnose those decisions and test specific changes without substituting new populations or requirements for the original ones.

RQ1—Representation value. Structural features increase recall over length alone by 64.07 percentage points internally (95% domain-clustered interval [59.99, 67.63]) and 76.81 externally (corrected independent-pair envelope [57.27, 87.19]). Internal structural FPR is 0.7747%, but external certified FPR rises to 90.5487%, versus 9.4513% for length-only. The zero-width cascade selects no evaluated transformer calls, so it adds no recall. Transformer-only external FPR is 99.2791%. H1 is unsupported: structural recall gain is real within the stated comparisons, but external specificity and incremental cascade benefit do not meet the joint requirement.

D's exact invariance across 8,622 pairs demonstrates a representation property while its refitted model loses 184 detections and removes 251 false positives. F, with weights and threshold fixed, removes 2,211 false positives by forcing HTTPS but loses 47 of 69 gold detections. R's wider gate adds 18 false positives without positive gain. Together, these results distinguish stable representation, useful ranking and low-FPR operation; improving one is not sufficient evidence of another.

RQ2—Monitoring and action. The GMM identifies 115/132 external windows (87.12%), exceeding the 80% sensitivity target. Its 28/252 reference alerts (11.11%)

exceed 5%. Future-only routing invokes the transformer on 94.8512% of the external stream, adds 218 certified false positives and leaves gold recall at 69/69. Policy FPR is 2479/2497 (99.2791%). H2 is unsupported because reference alerts, policy FPR and false-negative reduction do not pass. Detecting input-distribution departure is therefore demonstrated; beneficial remediation is not. There are no observed gold misses for this policy to repair in the evaluated stream.

RQ3—Inline tradeoffs. The original cascade records zero calls on the 10,000-request reference and zero errors in the 50,000 primary concurrency-64 attempts. Its pooled all-request p95 is 364.30 ms, above 200 ms. Throughput is approximately 1,244–1,272 requests/s at concurrency 8 but 399–503 at 64. External certified and Tranco alert rates exceed 1%; both systems alert on all 1,163 Tranco controls. The corrected 69-case paired lower bound of -6.1533 points does not establish the -2-point recall margin. These failures of joint conditions leave H3 unsupported despite call economy and primary request reliability.

The subsequent measurements explain what can be improved. S reduces median paired structural HTTP p95 by 79.68%, with agreement on all 99,999 comparable predictions, but not the full admission-order response contract. R separately supports two recall-only noninferiority statements on 1,392 positive domains under independent labeled-pair assumptions; its larger sample does not repair specificity or revise original H3. C measures 1.646 times structural computation for symmetric routing, with 0.223-ms p95 but 6.769-ms p99. Rare expert calls are cheap enough to escape p95 yet dominate the farther tail. These findings identify where additional computation and client organization matter, and where neither can correct shared detection errors.

5.2 End-to-End Argument and Contribution

The central engineering question is whether an additional computation repairs a decision that the cheaper model gets wrong. The original external outcomes contain 2,261 shared false positives among 2,497 certified negatives. No selector of those unchanged binary expert decisions can reduce observed FPR below 90.5487%, even if the selector knows the labels. This is an offline feasibility bound for those decisions, not an impossibility theorem for new features, thresholds or models. It explains why routing more of the same examples cannot solve that particular specificity problem. The bounded revised-rule experiment tests this reasoning beyond the original rows: on R, both selected gates introduce 18 false positives without repairing a structural error.

The representation comparisons identify a different intervention point. F establishes that scheme spelling can substantially alter frozen structural decisions, including both correct detections and false alarms. D removes the scheme pathway from every affected feature, rather than deleting only is_https, and verifies exact invariance. Its detection tradeoff shows that stable representation and low-FPR discrimination are separate requirements. Consequently, the appropriate design decision is to retain the tested conformance property as an input contract while requiring new labeled evidence before claiming that it improves deployment performance.

The computation and service comparisons locate two distinct costs. C measures real expert execution and shows that widening a cheap gate increases processing time without a detection return on the expanded cohort. S holds the structural scorer fixed and demonstrates a large client/service-path latency reduction through persistent client ownership. The results do not collapse into one generic speedup: one concerns work

performed by the detector, the other client-observed HTTP latency under a specified load and response contract. A systems designer can use this distinction to decide whether to change routing, representation or transport organization, rather than attributing every delay to the model.

This work extends an existing research discussion, rather than claiming priority for normalization, cross-source evaluation or calibrated cascades. Ahamed et al. (2026) already examine robustness and a separate PhishTank/Majestic evaluation; Rashid et al. (2024) and Tsai et al. (2024) address transfer and dataset bias. Li et al. (2021) and Alajaji (2026) establish cascade and selective-deferral precedents. ExpertFusion's author repository describes fixed URL encoders, lexical features, prior-specific experts and calibrated routers using unlabeled target statistics (aI-area, 2026; Hussain et al., 2027). The distinction established here is the measured connection among same-example expert repair, a fixed risk constraint, actual invocation cost and a separately controlled HTTP intervention. The original algorithms are established; the empirical contribution is the tested decision logic and its boundary conditions in this URL-only gateway.

The research scope remains the URL-only gateway in Sections 1.4–1.5. The original 22 checks and 25 operational groups answer the stated representation, monitoring and inline-execution questions. Later comparisons explain their mechanisms and test specific modifications under separate designs. In particular, the larger R cohort establishes recall-only noninferiority under its stated sampling assumptions; it neither revises the original 69-case result nor supplies new support for H2. No original threshold, model or joint decision is replaced after observing a follow-up outcome.

5.2.1 Technical Value of the Completed Investigation

Table 5.1 connects the findings to reusable design decisions: measure expert repair before widening a gate; distinguish invariant representation from low-FPR discrimination; assess the action following a monitor alert; and separate model computation from client-observed HTTP cost. The contribution is this tested connection among constraints, not a new component algorithm or the volume of retained artifacts.

Table 5.1. Technical contributions, supporting observations and reusable value.

Contribution	**Evidence in this study**	**Value to a systems researcher**
Joint requirements with valid paired uncertainty	All 22 original checks assessed; eight pass and fourteen do not. R separately supports recall-only noninferiority.	Separates a supported component or contrast from a joint operating claim and avoids zero-width population certainty.
Conditional expert-repair boundary	Original shared-error floor 90.5487%; R selected gates add 18 false positives and no detections.	Tests whether available expert decisions can solve the targeted errors before widening a gate.
Representation mechanism and conformance	F loses 47 gold detections while removing 2,211 false positives. D verifies exact invariance on 8,622 pairs.	Distinguishes spelling sensitivity, successful invariance and useful two-class detection.
Monitoring-to-action evaluation	115/132 external window alerts; 28/252 reference alerts; 218 added certified false positives.	Measures the utility of the action after an alarm instead of equating distribution sensitivity with remediation.

Contribution	Evidence in this study	Value to a systems researcher
Actual detector cost versus HTTP cost	C symmetric ratio 1.646 with expert-driven p99; S median paired p95 reduction 79.68% on structural synthetic requests.	Separates selective model work from client/service latency and makes the relevant timing boundary explicit.
Direct fit-input evidence	All five new shuffled-label fits have verified consumed-label linkage; every ranking outcome is retained.	Makes a negative-control implementation inspectable without inferring correctness from score appearance.

5.2.2 Engineering Work and Reusable Research Artifacts

The implemented gateway links source preparation, domain allocation, fitting, operating-point selection, inference, monitoring and HTTP execution to reconstructable observations. The original matrix retains all 1,243,505 measured attempts and 1,901 errors across 25 groups. Later comparisons retain every seed, input spelling, routing system and scheduled arm. Appendix A supplies the path from each research question to its table, aggregate and source lineage. Reuse requires a newly declared operating population and fresh evaluation; the present results are not guarantees for another detector or deployment.

5.2.3 What the Engineering Iteration Adds

Table 5.2 distinguishes original joint decisions from component and later-study outcomes. D establishes a representation property with a quantified detection tradeoff; F isolates spelling sensitivity under fixed weights. R tests whether selected expert use transfers beyond development; C measures its actual cost. S changes client organization

while holding the structural scorer fixed. Together these comparisons replace a generic claim of improvement with an identified intervention, its measured effect and the requirement it does or does not satisfy.

Table 5.2. Decisions at the hypothesis, component and follow-up levels.

Claim or requirement	**Evidence and achieved components**	**Overall decision**
Original H1: representation and low-FPR detection	Five of ten primary checks pass: three internal specificity checks and two positive structural-versus-length recall contrasts	Not supported; external specificity and incremental cascade gain do not meet the joint criteria
Original H2: monitoring and useful escalation	One of four primary checks passes: 115/132 external windows trigger alerts	Not supported; audit false alarms, policy specificity and incremental gold recall do not meet the joint criteria
Original H3: inline detection –service tradeoff	Two of eight primary checks pass: physical call limit and request-error limit	Not supported; external risk and pooled p95 requirements are not met, and the two-point recall margin is not established

Claim or requirement	Evidence and achieved components	Overall decision
Follow-up D: scheme-neutral detection	Exact invariance on 8,622 paired inputs; 251 fewer false positives and 184 fewer detected positives at frozen operating points	Not supported; recall-gain and low-FPR requirements are not met
Follow-up S: client-topology service comparison	Four of five requirements pass: median ratio 0.203249; interval upper 0.213508; worker p95 71.9990 ms; zero worker errors	Not supported as a conjunction; strict agreement is 799/100,000. All 99,999 comparable predictions agree
Post-review R: recall-only comparisons	1,392 concordant detections; conservative 95% envelope ±0.3143 points for original and symmetric versus transformer	Both supported conditional on independent labeled pairs; not original H3 or deployment adequacy

Claim or requirement	Evidence and achieved components	Overall decision
Post-review R: symmetric benefit and budget	21/2,654 logical calls; 18 added false positives; no added positive detections	Not supported: call limit passes, recall-gain and specificity conditions do not
Post-review F, C and P diagnostics	Controlled spelling effect; 30 actual computation arms; five fit-boundary-audited controls	Mechanism and integrity findings, not newly invented joint hypotheses

5.3 Implications for Practice

A deployment decision needs labeled negative evidence at the intended operating point: high positive recall alone can coexist with near-universal false alarms. Tranco offers a useful control but is not verified benign truth (Le Pochat et al., 2019). Routing should report membership, repairs, introduced errors and actual computation together. Monitoring should justify the action following an alarm under the same risk constraint. Stable representation must be evaluated on both classes.

Measure the actual service path, including transport, queueing and validation. Prediction equality does not imply equality of admission order or other stateful response

fields. Keep closed-loop throughput, request latency, physical attempts and post-timeout drain work distinct. The present single-host comparisons inform these design choices without establishing production arrival-rate capacity, cross-hardware transfer or suitability for automatic blocking.

5.4 Limitations and Threats to Interpretation

Publisher classifications define the outcomes; they are not new forensic labels. The original 69 gold-positive domains give the corrected ±6.1533-point paired envelope, while 2,497 certified negatives occupy only 234 domains. R's one-URL-per-domain cohort changes the estimand and narrows recall uncertainty, but distinct domains do not prove campaign independence. Its recall-only decisions are conditional on independent labeled pairs, and neither cohort estimates traffic prevalence.

Domain exclusion does not separate collection mechanisms. PhiUSIIL lacks independent per-row source identity, precluding a noncircular source-only classifier. F identifies a frozen-model spelling effect, not the complete cause of transfer failure. P proves consumed-label linkage for its five new fits; variable validation ranks, true-label threshold selection and the small permutation count prevent leakage-certification or formal significance claims. Historical fits retain their original audit boundary.

Information exposure matters. Publisher-url_norm adoption followed preparation and label-count inspection; checkpoint recovery followed predictions and partial measurement. Earlier analyst exposure and repeated scoring distinguish the study from a blind, single-lifetime-pass evaluation. Models and thresholds remain fixed, and the complete chronology is retained in Appendix C and its supplement. GMM windows overlap and routing activations span domains; saved-outcome resampling is conditional

on the observed order. Label-free probes do not establish semantic preservation or real-world adversarial robustness.

Operational results concern one host, fixed workload order and two original sessions. Session, thermal, cache and elapsed-time effects are not separately identified. Closed-loop timing excludes unsent backlog. D's retrospective publisher-labeled benchmark shares upstream mechanisms and conflicting collection-month descriptions; its domain exclusions do not establish temporal independence. Refitting and recalibration form a combined intervention, while scheme companions test representation rather than live-site equivalence.

S estimates client-topology effects for synthetic structural-only HTTP work and ten primary run pairs. C measures real-string detector computation at concurrency one with a designed 1% mixture, five cyclic rotations that are not fully position-balanced, and no run-level confidence interval. They cannot be pooled into a transformer-service speedup. Sampled host checks and saved-evidence verification establish recorded conditions and arithmetic, not disturbance-free operation or independent empirical replication.

5.5 Future Research, Separate from This Study

The measured shared-error floor motivates changed discriminative information or operating-point development, tested on independently adjudicated, time-indexed sources. More routing among the same decisions is not an adequate remedy. Operational follow-up should cross workload and session order, test open-loop arrivals and distinguish stable admission order from request-correlated prediction equality. These are new studies

suggested by the completed comparisons, not replacements for present thresholds or decisions.

5.6 Conclusion

The study answers RQ1–RQ3 through completed original measurements and mechanism-directed comparisons. Eight of 22 original component checks pass, but H1–H3 remain unsupported under their joint requirements. H1 combines a real structural recall gain with external specificity loss and no original cascade gain. H2 detects external distribution departure, but its routing action adds false positives without recovering a gold miss. H3 demonstrates call economy and primary request reliability while missing the original latency, external-risk and recall-margin requirements. The post-review R cohort separately establishes recall-only noninferiority to the transformer on 1,392 positive domains, conditional on the stated assumptions; it does not change those original decisions.

The modifications and diagnostics produce concrete engineering knowledge. D achieves exact invariance on 8,622 paired inputs while quantifying the detection cost of its refitted representation. F demonstrates why a scheme-prefix change that removes thousands of false alarms is still unacceptable when it loses 47 of 69 positive detections. R shows that the selected wider gate offers no measured external repair; C quantifies the additional work and exposes an expert-driven p99 tail hidden by the cheaper p95. P adds direct fit-boundary evidence for all five new shuffled-label controls. S demonstrates a 79.68% median paired p95 reduction for the tested structural HTTP workload, with prediction agreement on all 99,999 comparable pairs, while preserving its unmet strict-response requirement.

CyberSentinel makes these findings traceable from input representation through score, routing decision, physical model execution and HTTP response. The resulting contribution is a reproducible empirical account of when a component property becomes a useful system improvement and when another constraint prevents that inference. It gives a researcher specific measurements for deciding what to change next: conditional error repair before wider escalation, labeled operating-point performance alongside invariance, and actual detector and client-path costs rather than call counts alone. The narrative is therefore one of design, evaluation, diagnosis and measured comparison, with every favorable and adverse result retained in its proper scope.

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Appendix A—Evidence and Reproduction Map

The retained release research-record-2026-10-04, cited as Tallam (2026), supplies the original data and measured-code record. Table A.1 uses paths relative to dissertation/2026-10-04 and its aggregate-data directory. The 12 October revision adds a separate edition directory, dissertation/2026-10-12, with the corrected 22-check ledger, F/R/C/P aggregate evidence, advisor response and revised presentation. Its evidence map covers all 31 manuscript tables and four figures, distinguishing inherited from new evidence. Literature citations support external claims; repository locators identify this study's artifacts, not independent corroboration.

Table A.1. Research questions, complete evidence products and reading order.

Question or concern	Read in the manuscript	Aggregate or provenance artifact
Exact commitments and hypotheses	Sections 1.4–1.5	provenance/advisor-deck-scope-crosswalk-20261001.md; primary-gates.csv
RQ1 representation comparison	Sections 3.6–3.8, 4.2, 5.1	primary-results.json; paired-contrasts.csv; secondary-metrics.csv
RQ2 monitor and routing effects	Sections 3.9–3.11, 4.3, 5.1	external-monitor-windows.csv; external-psi-features.csv; complete-secondary-results.json
RQ3 service tradeoffs	Sections 3.12, 4.4, 5.1	operational-runs.csv (125 rows); operational-groups.csv (25 rows)
Every primary decision	Section 4.7	primary-gates.csv (22 rows); primary-results.json
Ranking, calibration and prevalence	Section 4.5	low-fpr-score-curves.csv; calibration-bins.csv; prevalence-projections.csv
Controls, sources, seeds and probes	Sections 4.5, 5.4	source-contingency.csv; seed-logical-invocations.csv; probe-decisions-and-scores.csv; probe-monitors-and-scores.csv
Integrity and methodological history	Sections 3.13, 4.6	verification.json; secondary-verification.json; provenance/Historical_Evidence_Supplement_2026-10-01.md

The original measured code revision is 77d128377ce5b401437d7179f5cd78fb4294b72c; the bounded follow-up revision is ef8ba5f0b357cf3dd60c4d663e6297d13334460c. The repository's research-archive/2026-10-04 guide supplies download, checksum, materialization and saved-observation

recomputation commands. Its catalog inventories licensed raw data, prepared splits, fitted artifacts, predictions, requests, freezes and interrupted attempts. Private execution capabilities and full host/process logs remain excluded or explicitly projected. Checksums establish identity, and recomputation checks arithmetic; neither constitutes new model fitting, timing or independent empirical replication.

Public method records are in provenance/followup, with detection results in aggregate-data/detection-D and complete service results in aggregate-data/service-S. Materialized follow-up inputs, manifests, model freezes and population admissions are under dissertation/followup-20261001 in the archive. The detection supplement retains two model rows, twenty calibration bins, two invariance rows and domain sizes. Its separate arithmetic implementation checks counts, rank-based AUC, average precision, Brier score and domain-bootstrap summaries. Candidate fitting used Python 3.10.19, NumPy 2.2.6, SciPy 1.15.3 and scikit-learn 1.7.2. Historical analysis-scripts retain controlled-workspace dependencies; the public recomputation adapter and data dictionaries specify the supported use.

Software checks have a separate chronology. The follow-up freeze recorded 215 focused passing tests while a broader run was in progress. That run finished with 13,016 passed, three failed and two skipped: a mock client lacked the async aclose method, preventing the intended SIGINT injection. A separately retained fixture-only correction passed all six original interruption cases, including three previously passing shift cases. The frozen source, original failure log and initial configuration-selection error before collection remain preserved. Current GitHub Actions results identify their tested

commits; they do not relabel historical failures or establish manuscript acceptance. No advisor or institutional approval is inferred from these checks.

Appendix B—Complete Follow-up Service Evidence

The following 80 rows preserve schedule order. Each arm has 1,000 warmups and 10,000 measured attempts. IDs identify workload, concurrency, pair and client; structural_detector is the unchanged structural scorer, and no_model is the response control. Successful-response p95 is reported here; the machine-readable arm file also provides successful, failed and all-request p50/p95/p99, measured/drain durations, error categories and physical counts. All eight pooled groups and all 40 pair-agreement rows accompany the package. Empty failure quantiles mean zero failures, not zero-millisecond failures.

Table B.1. Complete 80-arm schedule; success-only p95 in milliseconds, attempted throughput in requests/s and terminal error counts.

Arm ID	Success p95	Requests/s	Errors
no_model-c1-pair01-shared	1.1002	1041.13	0
no_model-c1-pair01-worker	1.1114	1037.84	0
no_model-c1-pair02-worker	1.1245	1019.56	0
no_model-c1-pair02-shared	1.0941	1041.69	0
no_model-c1-pair03-shared	1.1086	1035.64	0
no_model-c1-pair03-worker	1.0958	1041.49	0
no_model-c1-pair04-worker	1.1181	1032.21	0
no_model-c1-pair04-shared	1.1015	1040.88	0
no_model-c1-pair05-shared	1.1098	1035.75	0

Arm ID	Success p95	Requests/s	Errors
no_model-c1-pair05-worker	1.0990	1042.91	0
no_model-c1-pair06-worker	1.0963	1040.99	0
no_model-c1-pair06-shared	1.0975	1044.50	0
no_model-c1-pair07-shared	1.0971	1039.30	0
no_model-c1-pair07-worker	1.1045	1040.20	0
no_model-c1-pair08-worker	1.1310	1028.41	0
no_model-c1-pair08-shared	1.1080	1040.75	0
no_model-c1-pair09-shared	1.1011	1036.62	0
no_model-c1-pair09-worker	1.0941	1045.86	0
no_model-c1-pair10-worker	1.1017	1036.23	0
no_model-c1-pair10-shared	1.0979	1042.61	0
no_model-c64-pair01-shared	341.8227	470.09	0
no_model-c64-pair01-worker	66.5180	1652.64	0
no_model-c64-pair02-worker	66.2435	1650.60	0
no_model-c64-pair02-shared	350.9110	466.41	0
no_model-c64-pair03-shared	327.2478	487.55	0
no_model-c64-pair03-worker	65.3162	1653.86	0
no_model-c64-pair04-worker	69.3325	1618.14	0
no_model-c64-pair04-shared	362.3160	440.58	0

Arm ID	Success p95	Requests/s	Errors
no_model-c64-pair05-shared	352.0579	469.31	0
no_model-c64-pair05-worker	67.2167	1647.16	0
no_model-c64-pair06-worker	66.4319	1660.42	0
no_model-c64-pair06-shared	350.3035	461.06	0
no_model-c64-pair07-shared	332.1273	486.86	0
no_model-c64-pair07-worker	67.0119	1671.26	0
no_model-c64-pair08-worker	67.9225	1637.49	0
no_model-c64-pair08-shared	354.3856	466.84	0
no_model-c64-pair09-shared	335.7060	480.74	0
no_model-c64-pair09-worker	68.1728	1640.31	0
no_model-c64-pair10-worker	66.3344	1645.62	0
no_model-c64-pair10-shared	350.3558	466.32	0
structural_detector-c1-pair01-shared	1.4180	819.89	0
structural_detector-c1-pair01-worker	1.4179	821.63	0
structural_detector-c1-pair02-worker	1.4262	819.52	0
structural_detector-c1-pair02-shared	1.4287	821.01	0
structural_detector-c1-pair03-shared	1.4133	826.39	0
structural_detector-c1-pair03-worker	1.4262	824.31	0
structural_detector-c1-pair04-worker	1.4413	816.86	0

Arm ID	Success p95	Requests/s	Errors
structural_detector-c1-pair04-shared	1.4103	826.39	0
structural_detector-c1-pair05-shared	1.4393	814.60	0
structural_detector-c1-pair05-worker	1.4103	819.61	0
structural_detector-c1-pair06-worker	1.4283	819.98	0
structural_detector-c1-pair06-shared	1.4164	822.56	0
structural_detector-c1-pair07-shared	1.4487	816.54	0
structural_detector-c1-pair07-worker	1.4211	821.83	0
structural_detector-c1-pair08-worker	1.4415	817.97	0
structural_detector-c1-pair08-shared	1.4195	820.35	0
structural_detector-c1-pair09-shared	1.4238	814.64	0
structural_detector-c1-pair09-worker	1.4304	817.64	0
structural_detector-c1-pair10-worker	1.4218	818.32	0
structural_detector-c1-pair10-shared	1.4200	822.27	0
structural_detector-c64-pair01-shared	343.1552	467.91	0
structural_detector-c64-pair01-worker	72.6055	1517.21	0
structural_detector-c64-pair02-worker	67.2667	1575.63	0
structural_detector-c64-pair02-shared	371.7900	442.31	1
structural_detector-c64-pair03-shared	359.2445	456.39	0
structural_detector-c64-pair03-worker	70.0094	1533.45	0

Arm ID	Success p95	Requests/s	Errors
structural_detector-c64-pair04-worker	72.8568	1509.99	0
structural_detector-c64-pair04-shared	358.8575	450.21	0
structural_detector-c64-pair05-shared	397.6750	416.67	0
structural_detector-c64-pair05-worker	71.5949	1532.53	0
structural_detector-c64-pair06-worker	74.7204	1503.85	0
structural_detector-c64-pair06-shared	342.2972	472.64	0
structural_detector-c64-pair07-shared	362.0759	442.19	0
structural_detector-c64-pair07-worker	74.3130	1525.00	0
structural_detector-c64-pair08-worker	71.7555	1522.63	0
structural_detector-c64-pair08-shared	352.6524	464.16	0
structural_detector-c64-pair09-shared	341.7835	475.67	0
structural_detector-c64-pair09-worker	73.6319	1509.45	0
structural_detector-c64-pair10-worker	71.2943	1539.79	0
structural_detector-c64-pair10-shared	353.7333	459.72	0

The service source is the unchanged ef8ba5f0b357cf3dd60c4d663e6297d13334460c checkout. The recovery authorization binds the separate launcher, tests, amendment, original manifest and interrupted-attempt preservation receipt. The independent arithmetic verifier uses separately implemented sorted linear quantiles, medians, paired-response comparisons, physical-count reconciliation and the fixed PCG64 bootstrap. Its verification record binds the retained

sources and five aggregate CSVs. It is a second calculation, not an external replication. All failure observations, including the original power interruption, shared-client timeout and unmet strict agreement, are retained. No fitting, prediction or measurement is performed during document synthesis.

Appendix C—Methodological Chronology and Execution Record

The full chronological record and exact source, model and code identities are retained in Extended_Methods_and_Chronology_2026-10-12.docx and the versioned research package. This appendix summarizes changes affecting interpretation. The supplement preserves the prior detailed text verbatim; its descriptions of original methods remain historical, not replacements for the corrected analysis in Section 3.8.

Protocol v1.10 governs the research record. Model selection and validation fixed the accepted structural and character artifacts before evaluation. A failed liblinear predecessor led to the documented SAGA specification; scoring and durable-publication corrections remain separately identified. Group-test text displays established analyst exposure before scoring. The September 21 singleton-scoring amendment followed a development comparison with one changed transformer decision; historically selected operating points were carried forward without recalibration.

Initial external parsing found no eligible gold or Tranco population because publisher raw strings lacked schemes. Exact publisher url_norm was then adopted consistently for parsing and model input, with raw cells preserved. Preparation and label-count inspection preceded this amendment, whereas amended predictions followed it. Source mappings, overlap rules, models and hypothesis thresholds did not change.

Four interrupted attempts preceded the accepted operational completion. The September 30 completion directive established bounded recovery after predictions and partial measurement. An all-or-none review admitted cells 1–72 from the retained prefix, and the continuation measured cells 73–125 in order. Cell 73 was measured anew, not assembled across attempts. Original primary H3 cells remain within the first session;

descriptive group 71–75 crosses the boundary. Every failed predecessor remains separate from accepted evidence.

D separately changed representation, refitted and recalibrated on original development partitions, then evaluated the admitted retrospective benchmark. A preceding schedule ended at an environmental stop condition during S. A disclosed replacement rule preserved those nonprimary observations, required a complete new 80-arm schedule and prohibited pooling. Final S retains the unchanged structural scorer and all original response requirements, including its unmet strict-equivalence condition.

After the October 12 review, corrected paired inference changed one original component decision. P, F, R and C were then specified and executed once: five audited shuffled-label fits, frozen-model scheme interventions, domain-balanced retrospective routing comparisons and 30 actual-computation arms. Frozen research commit: f77027b3963f4e7ee6b831b9f09ff04c71491001. Execution-profile SHA256: 84e4d47f2c7c6249766dc677b6b7c2cad5be8fdaefb65195d786be418463c4d6. Independent saved-evidence verification checks 506 bound files and 258 recorded host observations, without new fitting or prediction. Host checks cover sampled power, thermal and known workload indicators, not every operating-system activity.

The revision preserves the original manuscript, failures, source evidence and numerical requirements. File identities establish provenance; they neither grant data access nor imply advisor approval or independent empirical replication. New raw URLs, domains, record identifiers and record-level predictions remain private. Aggregate results and permitted source identities support the public research record.